MATHEMATICS

CIRCLE
THEORY SHEET
ENGLISH

MATRIX

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1 BASIC GEOMETRY ABOUT CIRCLES

(i) Equal chords subtends equal angles at the centre and vice-versa.

(ii) Equal chords of a circle are equidistant from the centre and vice-versa.

(iii) Angle subtended by an arc at the centre is double the angle subtended at any point on the remaining part of the circle.

(iv) Angles in the same segment of a circle are equal.

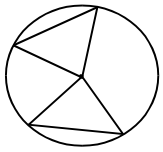


fig - (i)

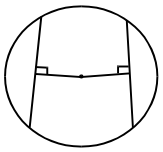


fig - (ii)

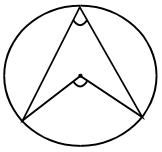


fig - (iii)

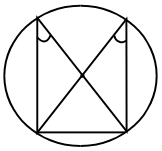


fig - (iv)

(v) The sum of the opposite angles of a cyclic quadrilateral is 180^0 and vice-versa.

(vi) If two chords of a circle intersect either inside or outside the circle at P then $AP \times PB = CP \times PD$

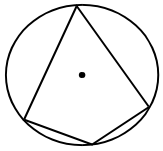


fig - (v)

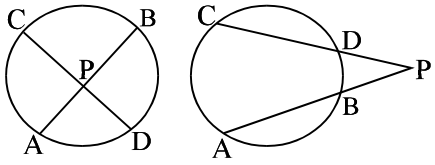


fig - (vi)

(vii) If a line touches a circle and from the point of contact a chord is drawn, the angles which this chord makes with the given line are equal respectively to the angles formed in the corresponding alternate segments.

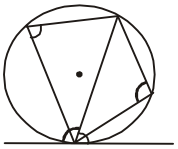


fig - (vii)

(viii) The greater of the two chords in a circle is nearer to the centre than lesser.

(ix) A chord drawn across the circular region divides it into parts each of which is called a segment of the circle.

(x) The tangents at the extremities of a chord of a circle are equal.

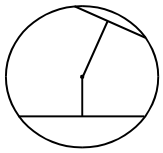


fig - (viii)

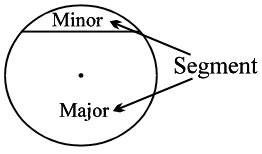


fig - (ix)

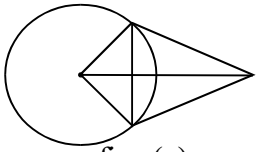


fig - (x)

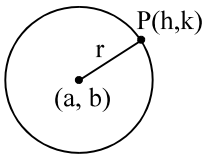
The angle between the tangents is bisected by the straight line, which joins their point of intersection to the centre. This straight line also bisects at right angles the chord, which joins the points where they touch the circle

2 CIRCLE : DEFINITION AND EQUATION

Definition :

A circle is the locus of a point which moves in a plane in such a way that its distance from a fixed point (in the same given plane) remains constant. The fixed point is called the centre of the circle and the constant distance is called the

radius of the circle.



$(h - a)^2 + (k - b)^2 = r^2$

$(x - a)^2 + (y - b)^2 = r^2$ ....(1)

fixed point is centre of the circle and fixed distance is radius of circle.

a, b ≡ centre

r ≡ radius

Note: If centre is origin then the equation of circle :

$x^2 + y^2 = r^2$

Expand (1) $x^2 + y^2 - 2ax - 2by + a^2 + b^2 - r^2 = 0$.

hence the general equation of the circle is taken as

$x^2 + y^2 + 2gx + 2fy + c = 0$. ....(2)

This can be written as $(x + g)^2 + (y + f)^2 = \left(\sqrt{g^2 + f^2 - c}\right)^2$

hence, Centre ≡ (–g, – f) i.e. $(-\frac{1}{2}$ coefficient of x ; $-\frac{1}{2}$ coefficient of y)

Radius ≡ $\sqrt{g^2 + f^2 - c}$

Remember that every second degree equation in x & y in which coeff. of x^2=coeff. of y^2 and there is no xy term represents a circle.

Note:

(i) Necessary and sufficient condition for general equation of degree 2 i.e. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ to represent a circle is

(a) coefficient of x^2 = coefficient of y^2 (not necessarily unity) and

(b) coefficient of xy = 0

Nature of circle :

$x^2 + y^2 + 2gx + 2y + c = 0$

where $r = \sqrt{g^2 + f^2 - c}$

If $g^2 + f^2 - c > 0$	If $g^2 + f^2 - c = 0$	If $g^2 + f^2 - c < 0$
⇒ Real circle with finite radius	⇒ point circle	⇒ imaginary circle

Questions on equation of circle

1. Find equation of circle

(i) Centre is (2,3) and circle passes through origin.

Ans. $x^2+y^2 - 4x - 6y = 0$

Sol. Centre is (2, 3) and circle passes through origin

$r = \sqrt{2^2 + 3^2} = \sqrt{13}$

Equation of circle : $(x - 2)^2 + (y - 3)^2 = 13$

$x^2 + y^2 - 4x - 6y = 0$

(ii) Centre is (1,1) and line 3x + 4y + 18 = 0 touches the circle.

Ans. $x^2 + y^2 - 2x - 2y - 23 = 0$

Sol. Centre is (1, 1) and line $3x + 4y + 18 = 0$ touches the circle.

$$r = \left|\frac{3\times1+4\times1+18}{\sqrt{3^2+4^2}}\right| = \frac{25}{5} = 5$$

Equation of circle : $(x-1)^2 + (y-1)^2 = 5^2$

$x^2 + 1 - 2x + y^2 + 1 - 2y = 25$

$x^2 + y^2 - 2x - 2y - 23 = 0$

(iii) Lines $x + y - 4 = 0$ and $2x + 3y - 8 = 0$ are diameters and area of circle is 25π.

Ans. $x^2 + y^2 - 8x - 9 = 0$

Sol. Lines $x + y - 4 = 0$ & $2x + 3y - 8 = 0$ are diameters and area of circle is 25π.

$x + y - 4 = 0$ ...(i), $2x + 3y - 8 = 0$ ...(ii)

from $2 \times$ (i) – (ii)

$2x + 2y - 8 = 0$

$2x + 3y - 8 = 0$

$- \quad - \quad +$

$-y = 0$

$y = 0$

$\Rightarrow x + 0 - 4 = 0$

$x = 4$

$\Rightarrow$ Centre = (4, 0)

$\because$ area = 25π

$\pi r^2 = 25\pi$

$r = 5$

then equation $(x-4)^2 + y^2 = 25$

$x^2 + 16 - 8x + y^2 - 25 = 0$

$x^2 + y^2 - 8x - 9 = 0$

(iv) Circumference is 16π and circle is concentric with $3x^2 + 3y^2 - 12x - 18y - 11 = 0$

Ans. $x^2 + y^2 - 4x - 6y - 51 = 0$

Sol. Circumference is 16π and circle is concentric with $3x^2 + 3y^2 - 12x - 18y - 11 = 0$

Centre of given circle : $(6, 9) \equiv (2,3)$

$\because$ Circumference = 16π

$2\pi r = 16\pi$

$r = 8$

then equation of circle

$(x-2)^2 + (y-3)^2 = 8^2$

$x^2 + 4 - 4x + y^2 + 9 - 6y = 64$

$x^2 + y^2 - 4x - 6y - 51 = 0$

(v) Circle passes through (6,0),(0,8) and (6,8)

Ans. $x^2 + y^2 - 6x - 8y = 0$

Sol. Circle passes through (6, 0), (0, 8) & (6, 8)

Let centre : (a, b)

$(a-6)^2 + b^2 = (a)^2 + (b-8)^2$

$a^2 + 36 - 12a + b^2 = a^2 + b^2 + 64 - 16b$

$-12a + 16b = 64 - 36$

$-4(3a - 4b) = 28$

$3a - 4b + 7 = 0$ ...(i)

& $(a-6)^2 + b^2 = (a-6)^2 + (b-8)^2$

$b^2 = b^2 + 64 - 16b$

$b = 4$

$\Rightarrow 3a - 16 + 7 = 0$

$a = 3$

Centre $\equiv (3, 4)$

$r = \sqrt{3^2 + 4^2} = 5$

Equation : $(x-3)^2 + (y-4)^2 = 5^2$

$x^2 + 9 - 6x + y^2 + 16 - 8y = 25$

$x^2 + y^2 - 6x - 8y = 0$

(vi) Circle passes through (0,0),(4,0) centre lies on line $y = x + 1$.

Ans. $x^2 + y^2 - 4x - 6y = 0$

Sol. Circle passes through (0, 0), (4, 0) centre lies on live $y = x + 1$

Let centre : $(\alpha, \alpha + 1)$

$\alpha^2 + (\alpha + 1)^2 = (\alpha - 4)^2 + (\alpha + 1)^2$

$\alpha^2 = \alpha^2 + 16 - 8\alpha$

$\alpha = 2$

$\therefore$ Centre = (2, 3)

$r = \sqrt{2^2 + 3^2} = \sqrt{13}$

$\Rightarrow$ Equation of circle :

$(x-2)^2 + (y-3)^2 = 13$

$x^2 + 4 - 4x + y^2 + 9 - 6y = 13$

$x^2 + y^2 - 4x - 6y = 0$

(vii) Passes through origin (0,0), radius is 5, and centre lies on x-axis.

Ans. $x^2 + y^2 + 10x = 0$ $\quad x^2 + y^2 - 10x = 0$

Sol. Let centre $(\alpha, 0)$

$\Rightarrow \sqrt{\alpha^2} = 5$

$\alpha = 5$ or -5

Centre : (5, 0) or -5

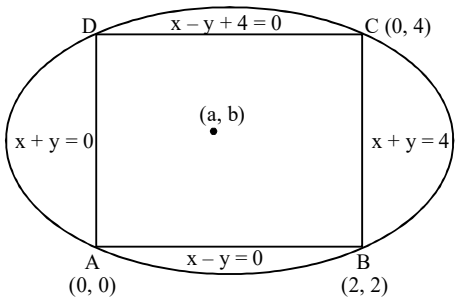
$\therefore$ Equation : $(x-5)^2 + y^2 = 5^2$ or $(x+5)^2 + y^2 = 5^2$

$x^2 + y^2 - 10x = 0$ $\quad x^2 + y^2 + 10x = 0$

(viii) Circumscribing the quadrilateral formed by lines $x - y = 0$, $x + y = 0$, $x - y + 4 = 0$, $x + y = 4$

Ans. $x^2 + y^2 - 4y = 0$

Sol.



Let centre : (a, b)

$a^2 + b^2 = (a-2)^2 + (b-2)^2$

$a^2 + b^2 = a^2 + 4 - 4a + b^2 + 4 - 4b$

$4a + 4b = 8$

$a + b = 2$ ...(i)

$a^2 + b^2 = a^2 + (b-4)^2$

$b^2 = b^2 + 16 - 8b$

$b = 2$

Centre : (0, 2)

$\Rightarrow a = 0$

$r = \sqrt{a^2 + b^2} = \sqrt{0+4} = 2$

Equation : $x^2 + (y-2)^2 = 2^2$

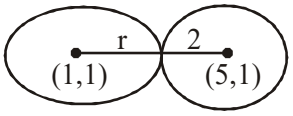
$x^2 + y^2 + 4 - 4y = 4$

$x^2 + y^2 - 4y = 0$

(ix) Centre is (1,1) and circle touches the circle $x^2 + y^2 - 10x - 2y + 22 = 0$ externally.

Ans. $x^2 + y^2 - 2x - 2y - 2 = 0$

Sol.



$\because r + 2 = \sqrt{(5-1)^2 + (1-1)^2}$

$r + 2 = 4$

$r = 2$

$\therefore$ Equation : $(x-1)^2 + (y-1)^2 = 4$

$x^2 + 1 - 2x + y^2 + 1 - 2y = 4$

$x^2 + y^2 - 2x - 2y - 2 = 0$

2. Prove that the points (3, 4), (– 3, – 4), (0, 5) and (–5, 0) are concyclic.

Sol. Let centre of a circle : (a, b)

$\Rightarrow (a-3)^2 + (b-4)^2 = (a+3)^2 + (b+4)^2$

$a^2 + 9 - 6a + b^2 + 16 - 8b = a^2 + 9 + 6a + b^2 + 16 + 8b$

$-12a = 16b$

$3a = -4b \Rightarrow a = -\frac{4}{3}b$

& $(a-3)^2 + (b-4)^2 = (a)^2 + (b-5)^2$

$a^2 + 9 - 6a + b^2 + 16 - 8b = a^2 + b^2 + 25 - 10b$

$6a + 8b = 10b$

$6a = 2b$

$3a = b$

Centre = (0, 0)

$r = \sqrt{3^2 + 4^2} = 5$

Equation of circle

$x^2 + y^2 = 25$ ...(i)

Point (–5, 0) satisfies the equation (i)

$\therefore$ Pts (3, 4), (–3, –4), (0, 5), (–5, 0) are concyclic.

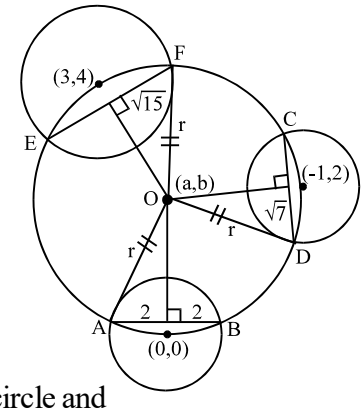
3. Find equation of circle passing through the extremities of the diameter of the circle $x^2 + y^2 = 4$; $x^2 + y^2 + 2x - 4y - 2 = 0$ and $x^2 + y^2 - 6x - 8y + 10 = 0$ **[O]**

ans. $(x-2)^2+(y-3)^2=17$

Sol. $r^2 = a^2+b^2+4$

$= (a+1)^2+(b-2)^2+7$

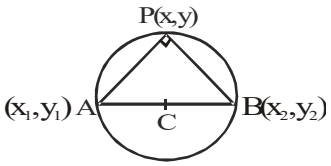
$= (a-3)^2+(b-4)^2+15$



3. DIAMETRICAL FORM OF A CIRCLE

If $A(x_1,y_1)$ and $B(x_2,y_2)$ are the end points of the diameter of the circle and

P(x,y) is the point other then A and B on the circle then from geometry



we know that $\angle APB = 90°$.

$\Rightarrow$ (Slope of PA) × (Slope of PB) = –1

$\Rightarrow \quad \therefore \left(\frac{y-y_1}{x-x_1}\right)\left(\frac{y-y_2}{x-x_2}\right) = -1$

$\Rightarrow \quad \mathbf{(x-x_1)\,(x-x_2)+(y-y_1)(y-y_2) = 0}$

Note : This will be the circle of least radius passing through (x_1, y_1) and (x_2, y_2)

$$\underbrace{(x-x_1)(x-x_2)}_{I} + \underbrace{(y-y_1)(y-y_2)}_{II} = 0$$

Diametrical form of circle is sum of 2 quadratic equation roots of equation I are abcissa of diametrical end pts. while roots of equation II are ordinates of diametrical end pts.

Questions on Diametrical Form of circle

1. The abscissa of 2 points 'A' and 'B' are the roots of the equation $x^2+2x-a^2=0$ and the ordinate are the roots of the equation $y^2+4y-b^2=0$. Find the equation of circle with AB as diameter.

Ans. $x^2+y^2+2x+4y-a^2-b^2=0$

Sol. $x^2+2x-a^2=0 \left\langle \begin{matrix} x_1 \\ x_2 \end{matrix}\right.$, $\qquad y^2+4y-b^2=0 \left\langle \begin{matrix} y_1 \\ y_2 \end{matrix}\right.$

can be written as $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$

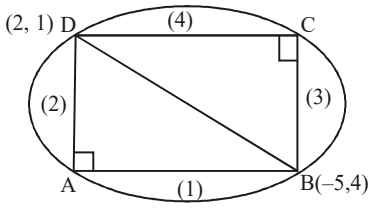
Coeff. of x^2 = Coeff. of y^2

2. Find the equation of a circle circumscribing the quadrilateral formed by the lines **[O]**

$2x+3y=2$; $3x-2y=4$; $x+2y=3$; $2x-y=3$

Ans. $8x^2+8y^2-25x-3y+18=0$

Sol.



$2x + 3y = 2$ ...(i)
$3x - 2y = 4$ ...(ii)
$x + 2y = 3$ ...(iii)
$2x - y = 3$ ...(iv)
$\because$ Line (1) $\perp$ (2) and line (3) $\perp$ (4)
$\therefore$ BD is diameter.
from (2) $\times$ (iii) – (i)
$2x + 4y = 6$
$2x + 3y = 2$
– – –
$y = 4$
$\Rightarrow x = 5 \Rightarrow B(-5, 4)$
from (ii) – 2 $\times$ (iv)
$3x - 2y = 4$
$4x - 2y = 6$
– + –
$x = 2$
$\Rightarrow y = 1$
$\therefore$ Equation of circle
$(x + 5)(x - 2) + (y - 4)(y - 1) = 0$
$(x^2 + 3x - 10) + (y^2 - 5y + 4) = 0$
$x^2 + y^2 + 3x - 5y - 6 = 0$

3. Line $y = x$ cuts the circle $x^2 + y^2 + 10x - 12y - 40 = 0$ at A and B. Find equation of the circle with AB as diameter.

ans. $x^2 + y^2 - x - y - 40 = 0$

Sol. $x^2 + y^2 + 10x - 12y - 40 = 0$
Put $y = x$
$x^2 + x^2 + 10x - 12x - 40 = 0$
$2x^2 - 2x - 40 = 0$

$x^2 - x - 20 = 0$ ...(i) $\begin{cases} x_1 \\ x_2 \end{cases}$

Put $x = y \Rightarrow y^2 - y - 20 = 0$ ...(ii) $\begin{cases} y_1 \\ y_2 \end{cases}$

Equation of circle
(i) + (ii) = 0
$x^2 - x - 20 + y^2 - y - 20 = 0$
$x^2 + y^2 - x - y - 40 = 0$

4. Line $y = mx + c$ cuts the curve $y^2 = 4ax$ at A and B. Find the equation of circle with AB as diameter.

ans. $m^2\left(x^2 + y^2\right) + 2cmx - 4ax - 4amy + 4acm + c^2 = 0$

Sol. Line $y = mx + c$, $y^2 = 4ax$ ...(i)
Put $y = mx + c$ in (i)
$(mx + c)^2 = 4ax$
$m^2x^2 + 2mcx + c^2 - 4ax = 0$

$m^2x^2 + (2mc - 4a)x + c^2 = 0 \left\langle \begin{matrix} x_1 \\ x_2 \end{matrix} \right.$ ...(ii)

& put $x = \left(\frac{y-c}{m} \right)$ in (i)

$y^2 = \frac{4a\cdot(y-c)}{m}$

× by m^2

$m^2y^2 = 4am(y - c)$

$m^2y^2 - 4amy + 4amc = 0 \left\langle \begin{matrix} y_1 \\ y_2 \end{matrix} \right.$ ...(iii)

From (ii) + (iii)

Equation of circle

$m^2(x^2 + y^2) + (2mc - 4a)x - 4amy + 4amc + c^2 = 0$

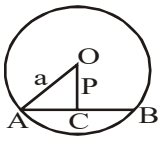
Note

All the circles passing through 2 fixed pts. A & B, smallest circle will be that circle for which A & B are diametrical end pts.

4. INTERCEPTS OF A CIRCLE

Intercept cut by a line on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ or

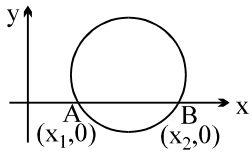
length of chord of the circle $= 2\sqrt{r^2 - p^2}$



where r is the radius and p is the length of perpendicular from the centre to the chord.

$x^2 + y^2 + 2gx + 2fy + c = 0$

x-intercept :

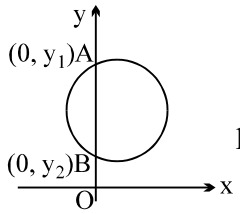


put $y = 0$

$x^2 + 2gx + c = 0$; $x_1 + x_2 = -2g$; $x_1 x_2 = c$

$|x_1 - x_2| = 2\left|\sqrt{g^2 - c}\right|$

y-intercept :

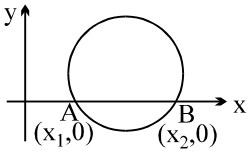


put x = 0

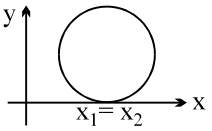
$y^2 + 2fx + c = 0$ roots are y_1 and y_2 $\quad |y_1 - y_2| = 2\left|\sqrt{f^2 - c}\right|$

Note:

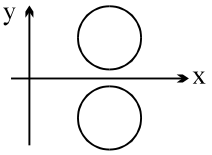
1. If $g^2 - c > 0$
   $\Rightarrow$ circle cuts the x-axis at 2 distinct points.



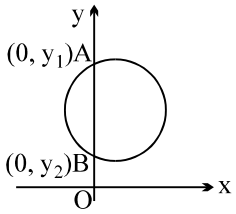
2. If $g^2 - c = 0$
   $\Rightarrow$ circle touches the x-axis
   In this case, y–coordinate of center will be +r or –r



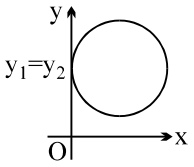
3. If $g^2 < c$
   $\Rightarrow$ circle lies completely above or below the x-axis.



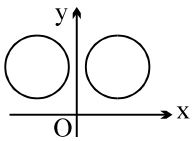
4. If $f^2 - c > 0$
   $\Rightarrow$ circle cuts the y-axis at 2 distinct points.



5. If $f^2 = c$
   $\Rightarrow$ circle touches the y-axis
   x–coordinate of center will be +r or –r



6. If $f^2 < c$
   $\Rightarrow$ circle lies completely either on right or on left of y-axis.

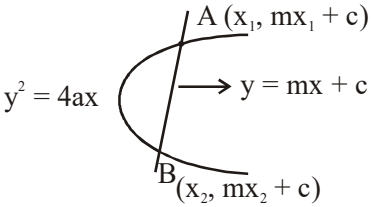


Questions on Intercepts

1. Find length of intercept cut by parabola $y^2 = 4ax$ on the line $y = mx + c$.

Ans. length $= |x_1 - x_2|\sqrt{1 + m^2}$

Sol.



$$AB = \sqrt{(x_1 - x_2)^2 + (mx_1 + c - mx_2 - c)^2}$$

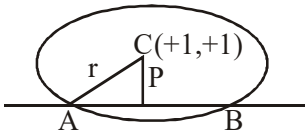
$$= \sqrt{(x_1 - x_2)^2 + m^2 (x_1 - x_2)^2}$$

$$|x_1 - x_2| \cdot \sqrt{1 + m^2}$$

2. Find length of intercept cut by $3x - 4y + 6 = 0$ on circle $x^2 + y^2 - 2x - 2y - 2 = 0$

Ans. $2\sqrt{3}$

Sol.



C (+1, +1)

$$r = \sqrt{1+1+2} = 2$$

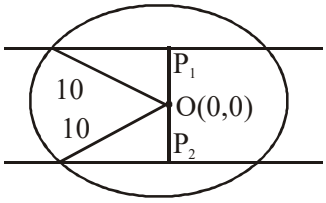
$$P = \left|\frac{3-4+6}{\sqrt{3^2+4^2}}\right| = \left|\frac{5}{5}\right| = 1$$

$$AB = 2\sqrt{r^2 - P^2} = 2\sqrt{4-1} = 2\sqrt{3}$$

3. If lines $3x - 4y + 10 = 0$ and $7x + 24y + c = 0$ cuts intercepts of the same length on circle $x^2 + y^2 = 100$ find c.

Ans. $c = \pm 50$

Sol.



$$P_1 = \left|\frac{10}{5}\right|, P_2 = \left|\frac{C}{25}\right|$$

$\because P_1 = P_2$

$$\frac{C}{25} = \pm 2$$

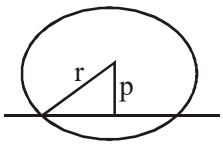
$C = \pm 50$

4. If lines $3x + 4y = c$ cut the circle $x^2 + y^2 = 100$ at two distinct points then find range of c.

Ans. $-50 < c < 50$

Sol. $r = 10$

$$p = \left|\frac{c}{5}\right|$$



$\because p < r$

$$\left|\frac{c}{5}\right| < 10$$

$-10 < \frac{c}{5} < 10$

$c \in (-50, 50)$

5. Find the equation of a circle which touches the coordinate axes and whose radius = 2.

Ans $(x \pm 2)^2 + (y \pm 2)^2 = 2^2$

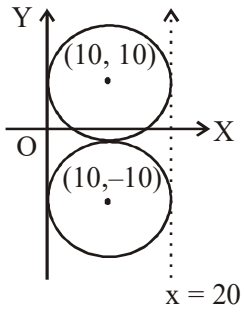
[Sol. 4 circles are possible

$(x \pm 2)^2 + (y \pm 2)^2 = 2^2$

6. Find equation of circle which touches the line x = 0, y = 0, x = 20

Ans. $x^2 + y^2 - 20x \pm 20y + 100 = 0$

Sol.



Equation of circle

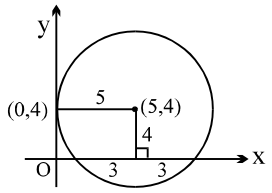
$(x - 10)^2 + (y \pm 10)^2 = 10^2$

$\Rightarrow x^2 + y^2 - 20x \pm 20y + 100 = 0$

7. Find the equation of a circle which touches the positive axis of y at a distance of 4 units from origin and cuts off an intercept of 6 units from the positive axis of x.

Ans. $(x-5)^2 + (y-4)^2 = 25$

Sol. From figure



equation of circle

$(x - 5)^2 + (y - 4)^2 = 5^2$

or $(x + 5)^2 + (y - 4)^2 = 25^2$

Aliter : Let the equation of circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$

passes through (0, 4) $\Rightarrow$ $16 + 8f + c = 0$ ....(1)

x-intercept = 6 $\Rightarrow$ $2\sqrt{g^2 - c} = 6$ $\Rightarrow$ $g^2 - c = 9$ ....(2)

Also touches the y-axis $\Rightarrow$ $f^2 = c$ ....(3)

Now proceed.

8. Find the equation of the locus of the centre of a circle which touches the positive y-axis and having intercept on x-axis equals to $2l$.

Ans $x^2 - y^2 = l^2$

Sol. from figure

$h^2 = k^2 + l^2$

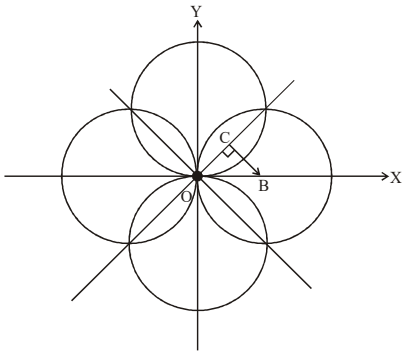
$\therefore$ required locus is

$x^2 - y^2 = l^2$

9. Find the equation of a circle passing through origin cutting off intercept equals to $\sqrt{2}$ on the lines $y^2 - x^2 = 0$.

Ans. $x^2 + y^2 \pm 2x = 0$; $x^2 + y^2 \pm 2y = 0$

Sol.



Centres will lie on x–axis or y-axis

If B is centre

$\therefore$ BC $\perp$ OA In ΔOCB

$OA = \sqrt{2} \Rightarrow OC = \dfrac{1}{\sqrt{2}}$ $\angle COB = 45^o = \angle OBC$

$OB^2\left(\dfrac{1}{\sqrt{2}}\right)^2 + \left(\dfrac{1}{\sqrt{2}}\right)^2 = 1$

$OB = \pm 1$

Centre $(\pm 1, 0)$, $r = 1$

If centre lie on Y-axis

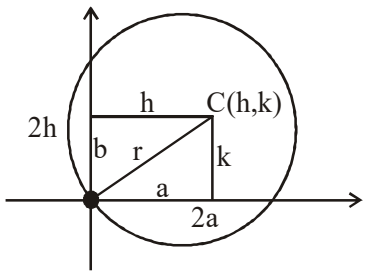
Centre $= (0, \pm 1)$, $r = 1$

Equation of circle	OR
$(x \pm 1)^2 + (y - 0)^2 = 1^2$	$(x - 0)^2 + (y \pm 1)^2 = 1^2$
$x^2 + y^2 \pm 2x = 0$	$x^2 + y^2 \pm 2y = 0$

10. Two rods whose lengths are 2a and 2b slide along the rectangular axes in such a way that their extremities are always concyclic. Find the equation of the locus of the centre of the circle.

Ans. $x^2 - y^2 = a^2 - b^2$

Sol.



$r^2 = a^2 + k^2$ ...(i)

and $r^2 = b^2 + h^2$ ...(ii)

from (i) & (ii)

$a^2 + k^2 = b^2 + h^2$

$h^2 - k^2 = a^2 - b^2$

$x^2 - y^2 = a^2 - b^2$

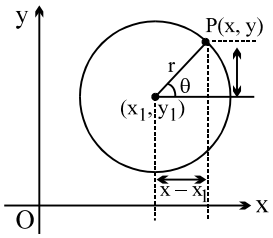
5. PARAMETRIC FORM OF A CIRCLE

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$$

$$x = x_1 + r\cos\theta$$

and $y = y_1 + r\sin\theta$

$x_1\, y_1 \longrightarrow$ fixed centre



$r \longrightarrow$ fixed radius and $\theta \in [0, 2\pi)$ is a parameter.

coordinates of any point on the circle is $x_1 + r\cos\theta$, $y_1 + r\sin\theta$
if $(x_1, y_1) \equiv (0, 0) \Rightarrow x = r\cos\theta$ & $y = r\sin\theta$
which are the parametric coordinates for the circle $x^2 + y^2 = r^2$
Note: If θ is eliminated we get Cartesian form of a circle i.e. $(x - x_1)^2 + (y - y_1)^2 = r^2$.

Questions on Parametric Form of a Circle

1. $x^2 + y^2 - 6x + 4y - 3 = 0$. convert into parametric form

Ans. $x = 3 + 4\cos\theta; y = -2 + 4\sin\theta$

Sol. $x^2 + y^2 - 6x + 4y - 3 = 0$ convert into parametric form
$g = -3, f = 2, c = -3$
Centre : $(3, -2)$

$$r = \sqrt{g^2 + f^2 - c} = \sqrt{9 + 4 + 3} = \sqrt{16} = 4$$

$P(x_1 + r\cos\theta, y_1 + r\sin\theta) \equiv (3 + 4\cos\theta, -2 + 4\sin\theta)$

2. If (a,b) lies on the circle $x^2 + y^2 - 2x - 2y - 2 = 0$, find maximum value of $5a - 12b$.

Ans. 19

Sol. $C(1,1)$, $r = \sqrt{1+1+2} = 2$
$(a, b) = (1 + 2\cos\theta, 1 + 2\sin\theta)$
$\Rightarrow 5a - 12b = 5(1 + 2\cos\theta) - 12(1 + 2\sin\theta)$
$= 5 + 10\cos\theta - 12 + 24\sin\theta$
$= -7 + (10\cos\theta + 24\sin\theta)$

$$\text{max} = \sqrt{10^2 + 24^2} = 26$$

$= -7 + 26 = 19$

3. Point P is variable point lying on the circle $x^2 + y^2 - 2x - 2y - 2 = 0$, find locus of mid point of OP.

Ans. $\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = 1$

Sol. Centre : $(1, 1)$, $r = \sqrt{1+1+2} = 2$
$P(1 + 2\cos\theta, 1 + 2\sin\theta)$, $O(0, 0)$
Let Q(h, k) is mid point of OP.

$$\therefore h = \frac{1 + 2\cos\theta}{2}, k = \frac{1 + 2\sin\theta}{2}$$

$$\cos\theta = \left(\frac{2h - 1}{2}\right), \sin\theta = \left(\frac{2k - 1}{2}\right)$$

$\because \cos^2\theta + \sin^2\theta = 1$

$$\left(\frac{2h-1}{2}\right)^2 + \left(\frac{2k-1}{2}\right)^2 = 1$$

$4h^2 + 1 - 4h + 4k^2 + 1 - 4k = 4$

$4h^2 + 4k^2 - 4h - 4k - 2 = 0$

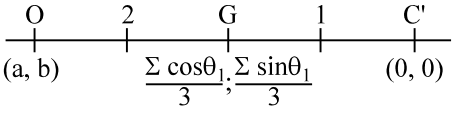
$$h^2 + k^2 - h - k - \frac{1}{2} = 0$$

$$x^2 + y^2 - x - y - \frac{1}{2} = 0$$

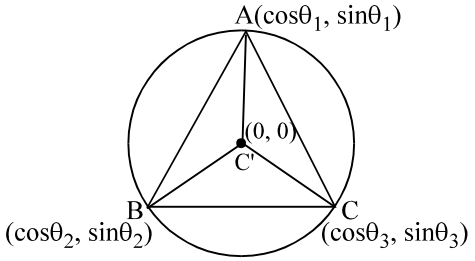
4. If $A(\cos\theta_1, \sin\theta_1)$; $B(\cos\theta_2, \sin\theta_2)$; $C(\cos\theta_3, \sin\theta_3)$ are the vertices of a triangle ABC then find the orthocenter of the triangle.

Ans $(\Sigma \cos\theta_1 ; \Sigma \sin\theta_1)$

Sol. A, B, C lies on a circle $x^2 + y^2 = 1$



$\therefore$ $O \equiv (\Sigma \cos\theta_1 ; \Sigma \sin\theta_1)$



5. 'P' is the variable point on the circle with centre at C. CA & CB are perpendicular from C on x-axis and y-axis respectively. Show that the locus of the centroid of triangle PAB is a circle with centre at the centroid of triangle CAB and radius equal to the third of the radius of the given circle.

Sol. Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

Let the centroid of the Δ CAB be (h, k)

$3h = x_1 - g + r\cos\theta$ }

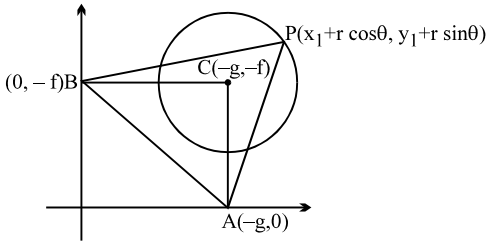
and $3k = y_1 - f + r\sin\theta$ }

where $x_1 = -g$; $y_1 = -f$

$\therefore$ $3h = -2g + r\cos\theta$

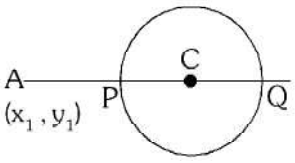
$3k = -2f + r\sin\theta$

$\Rightarrow$ $(3h + 2g)^2 + (3k + 2f)^2 = r^2$ Now proceed



6. POSITION OF A POINT WRT A CIRCLE

P lies outside the circle



$CP > r \Rightarrow (CP)^2 > r^2 \Rightarrow (x_1 + g)^2 + (y_1 + f)^2 > g^2 + f^2 - c$

$x_1^2 + 2gx_1 + g^2 + y_1^2 + 2fy_1 + f^2 > g^2 + f^2 - c$

$x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c > 0 \Rightarrow$ $S_1 > 0$

Note:

1. If P lies outside the circle, $S_1 > 0$
2. If P lies on the circle, $S_1 = 0$
3. If P lies inside the circle, $S_1 < 0$

	$x^2+y^2=4$	$y^2=8x$	$x^2-6xy+4y^2=8x$
$S\equiv$	x^2+y^2-4	y^2-8x	$x^2-6xy+4y^2-8x$
$S_1\equiv$	$x_1^2+y_1^2-4$	$y_1^2-8x_1$	$x_1^2-6x_1y_1+4y_1^2-8x_1$
$T\equiv$	xx_1+yy_1-4	$yy_1-4(x+x_1)$	$xx_1-3(xy_1+yx_1)+4yy_1-4(x+x_1)$

T can be obtained by replacing.

$x^2 \to x\,x_1$; $y^2 \to y\,y_1$; $2x \to x + x_1$; $2y \to y + y_1$; $2xy \to xy_1 + yx_1$

Questions on Position of a point w.r.t. a circle :

1. Determine position of point P(6, –7) and Q (9,8) with respect to circle $x^2 + y^2 = 100$.

Ans. P lies inside the circle. Q lies outside the circle

Sol. $x^2 + y^2 = 100$; $r = 10$

$S \equiv x^2 + y^2 - 100$

For P

$S_1 \equiv 6^2 + (-7)^2 - 100 = 36 + 49 - 100 = -15 < 0$

P lies inside the circle.

For Q

$S_1 \equiv 9^2 + 8^2 - 100 = 81 + 64 - 100 > 0$

Q lie outside the circle.

2. $S_1 = x^2 + y^2 - 4x + 6y - 3 = 0$

$S_2 = x^2 + y^2 + 4x - 6y - 3 = 0$

point (1, 2) lies

(A) inside $S_1 = 0$ and inside $S_2 = 0$ (B) outside $S_1 = 0$ and outside $S_2 = 0$

(C) inside $S_1 = 0$ and outside $S_2 = 0$ (D) outside $S_1 = 0$ and inside $S_2 = 0$

Ans D

Sol. $S_{11} = 1^2 + 2^2 - 4 \times 1 + 6 \times 2 - 3$

$= 1 + 4 - 4 + 12 - 3$

$= 10 > 0$

Outside $S_1 = 0$

$S_{21} = 1^2 + 2^2 + 4 \times 1 - 6 \times 2 - 3$

$= 1 + 4 + 4 - 12 - 3$

$= 9 - 15$

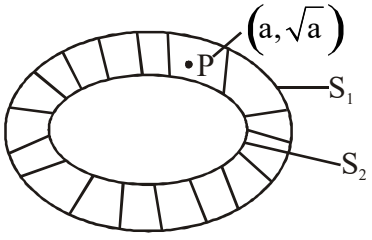
$= -6 < 0$

Inside $S_2 = 0$

3. If point P $\left(a, \sqrt{a}\right)$ lies in the area bounded by $x^2 + y^2 = 42$ and $x^2 + y^2 = 20$ then find a.

Ans. $a \in (4,6)$

Sol.



$S : x^2 + y^2 - 42;\ a > 0$

$\because P\left(a, \sqrt{a}\right)$ lies inside S_1.

$\therefore a^2 + a - 42 < 0$

$a^2 + 7a - 6a - 42 < 0$

$(a + 7)(a - 6) < 0$

$a \in (-7, 6)$

& $P\left(a, \sqrt{a}\right)$ lies outside S_2

$\therefore a^2 + a - 20 > 0$

$(a + 5)(a - 4) > 0$

$a \in (-\infty, -5) \cup (4, \infty)$

$\Rightarrow (a \in (4, 6))$

4. If P (a, 2a) lies in region between circles $x^2 + y^2 = 9$ and $x + y = 3$ in I quadrant then find a.

Ans. $a \in \left(1, \frac{3}{\sqrt{5}}\right)$

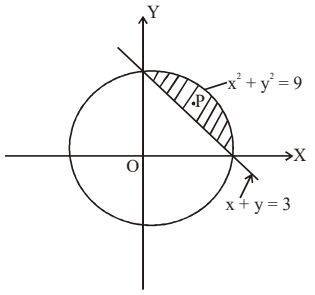
Sol. O(0, 0) & P(a, 2a) lie opposite of line $x + y - 3 = 0$

$\therefore (-3) \cdot (a + 2a - 3) < 0$

$(3a - 3) > 0$

$a > 1$

$a \in (1, \infty)$



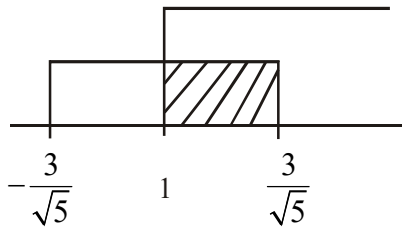
Point P(a, 2a) lies in side of circle

$\therefore a2 + 4a^2 - 9 < 0$

$5a^2 - 9 < 0$

$\left(a + \frac{3}{\sqrt{5}}\right)\left(a - \frac{3}{\sqrt{5}}\right) < 0$

$a \in \left(-\frac{3}{\sqrt{5}}, \frac{3}{\sqrt{5}}\right)$



$a \in \left(1, \frac{3}{\sqrt{5}}\right)$

5. If the line segment joining (x_1, y_1) & (x_2, y_2) makes an obtuse angle at (x_3, y_3) then prove that $(x_3 - x_1)(x_3 - x_2) + (y_3 - y_1)(y_3 - y_2) < 0$

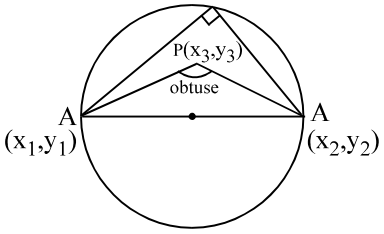
Sol. Equation of circle with AB as diameter

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

since AB subtends obtuse angle at (x_3, y_3)

$\therefore$ P lies inside the circle

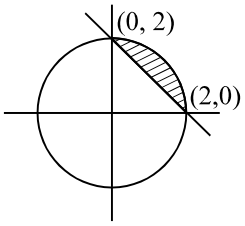
$\therefore$ $(x_3 - x_1)(x_3 - x_2) + (y_3 - y_1)(y_3 - y_2) < 0$ H.P.



6. For what value of a the point (a, a + 1) lies inside the region bounded by the circle $x^2 + y^2 = 4$ and the line $x + y = 2$ in the first quadrant.

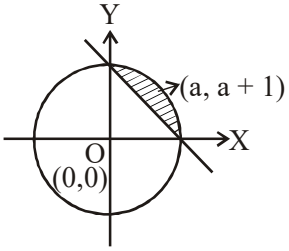
Ans $\left(\frac{1}{2}, \frac{\sqrt{7}-1}{2}\right)$

Sol



a will lie in $\left(\frac{1}{2}, \frac{\sqrt{7}-1}{2}\right)$

Alternate :



$\because$ O(0, 0) & P(a, a + 1) lies opposite side of line $x + y - 2 = 0$

$\therefore (-2)\cdot(a + a + 1 - 2) < 0$

$(2a - 1) > 0$

$a > \frac{1}{2}$

$a \in \left(\frac{1}{2}, +\infty\right)$

P lies inside of circle $x^2 + y^2 - 4 = 0$

$\therefore a^2 + (a + 1)^2 - 4 < 0$

$2a^2 + 2a - 3 < 0$

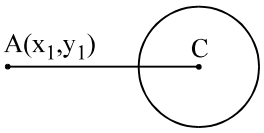
$$\left(a-\frac{\left(-1-\sqrt{7}\right)}{2}\right)\left(a-\left(\frac{\sqrt{7}-1}{2}\right)\right)<0$$

$$a\in\left(\frac{-1-\sqrt{7}}{2},\frac{\sqrt{7}-1}{2}\right)$$

$$\Rightarrow a\in\left(\frac{1}{2},\frac{\sqrt{7}-1}{2}\right)$$

Note:

1. Greatest and least distance of a point $A(x_1, y_1)$ from a circle with centre 'C' and radius 'r' is $|AC + r|$ and $|AC - r|$

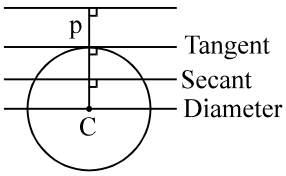


2. Distance of a line from a circle = $p - r$
where p = distance of line from center, r = radius of circle

7. LINE AND CIRCLE

Let L = 0 be a line & S = 0 be a circle. If r is the radius of the circle & p is the length of the perpendicular from the centre on the line, then:

(i) $p > r \Leftrightarrow$ the line does not meet the circle i. e. passes out side the circle.
(ii) $p = r \Leftrightarrow$ the line touches the circle. (It is tangent to the circle)
(iii) $p < r \Leftrightarrow$ the line is a secant of the circle.
(iv) $p = 0 \Leftrightarrow$ the line is a diameter of the circle.



General method

$$x^2 + y^2 = a^2$$

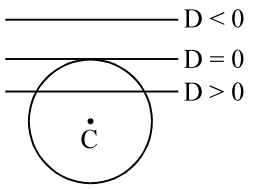
$$y = mx + c$$

Eliminate y

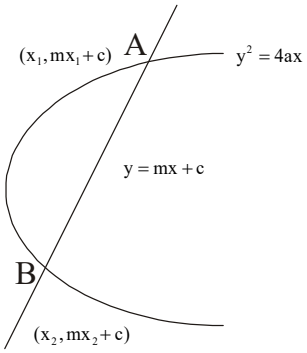
$$x^2 + (mx + c)^2 = a^2$$

quad. in x get D

(i) $D > 0 \Rightarrow$ line is a Secant.
(ii) $D = 0 \Rightarrow$ line is a Tangent.
(iii) $D < 0 \Rightarrow$ line passes outside the circle.



General Method for length of chord



$(mx + c)^2 = 4\ ax$

$m^2x^2 + x(2\ cm - 4a) + c^2 = 0$; x_1 and x_2 are its roots.....

get $|x_1 - x_2|$

$$\ell_{AB} = \sqrt{(x_1 - x_2)^2 + (mx_1 - mx_2)^2}$$

$$= \sqrt{(x_1 - x_2)^2 (1 + m^2)}$$

$$= |x_1 - x_2| \sqrt{1 + m^2}$$

8.1 TANGENT & NORMAL

Tangent

Tangent is a limiting case of secant

Condition of Tangency(COT)

The line L = 0 touches the circle S = 0 if p is the length of the perpendicular from the centre to that line and radius of the circle r are equal i.e. p = r.

Note:

1. If point of tangency is given then equation of tangent will be T = 0.
2. If any other information is given then we will assume equation of tangent & then apply COT.

For Circle $x^2 + y^2 = a^2$

1. Point of tangency is $P(x_1, y_1)$
   Equation of tangent $xx_1 + yy_1 = a^2$ (Cartesian form)
2. Parametric form : $x_1 = a\cos\theta, y_1 = a\sin\theta$
   $\therefore$ equation of tangent is $x\cos\alpha + y\sin\alpha = a$.
3. Slope form : $y = mx + c$ is tangent to the circle $x^2 + y^2 = a^2$
   if $c^2 = a^2(1 + m^2) \Rightarrow$ condition of tangency.

   $\therefore$ $y = mx \pm a\sqrt{1 + m^2}$ is the equation of the tangent to $x^2 + y^2 = a^2$.

Normal

Normal at a point is the straight line which is perpendicular to the tangent at the point of contact.

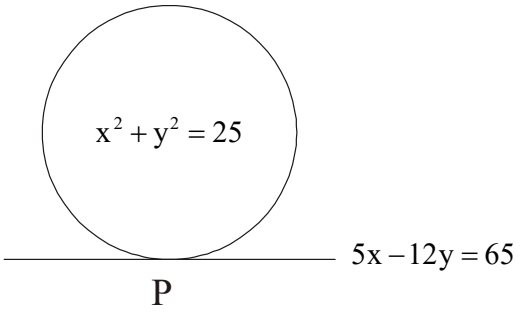
Normal at any point on the circle passes through the centre of the circle.

8.2 TO FIND POINT OF TANGENCY

Method – 1

Foot of perpendicular from center on tangent is point of contact of the tangent

Method – 2



let $P(x_1, y_1)$

equation of tangent in terms of x_1 & y_1

$T = 0$

$xx_1 + yy_1 = 25$

$5x - 12y = 65$

$$\frac{x_1}{5} = \frac{y_1}{-12} = \frac{25}{65} \Rightarrow \qquad x_1 = \frac{25}{13}, \quad y_1 = \frac{-60}{13}$$

Questions on Tangent and Normal :

1. For circle $x^2 + y^2 = 25$

(i) Find equation of normal passing through (2,1)

Ans. $x = 2y$

Sol. $(y - 0) = \frac{1}{2}(x - 0)$

$x = 2y$

(ii) Find equation of normal whose slope is $\frac{-2}{3}$

Ans. $2x + 3y = 0$

Sol. $(y - 0) = -\frac{2}{3}(x - 0)$

$3y + 2x = 0$

(iii) Find equation of tangent at $(4, -3)$

Ans. $4x - 3y = 25$

Sol. $T = 0$

$xx_1 + yy_1 - 25 = 0$ ...(i)

passes through $(4, -3)$

$4x - 3y - 25 = 0$

(iv) Find equation of tangent whose slope is –3.

Ans. $3x+y = \pm\sqrt{10}(5)$

Sol. $m = -3$

Tangent $y = -3x + c$ ...(i)

$\because c = \pm a\sqrt{1+m^2}$

$c = \pm 5\sqrt{1+9} = \pm 5\sqrt{10}$

Equation of tangent : $y = -3x \pm 5\sqrt{10}$

$3x + y = \pm 5\left(\sqrt{10}\right)$

(v) Find equation of tangent parallel to the line $5x - 12y = 0$

Ans. $5x - 12y \pm 65 = 0$

Sol. $m = \frac{5}{12}$

tangent $\Rightarrow y = mx + c$

$y = \frac{5}{12}x + c$ ...(i)

$\because c = \pm a\sqrt{1+m^2}$

$c = \pm 5\sqrt{1+\frac{25}{144}}$

$c = \pm \frac{5}{12}\sqrt{169} = \pm \frac{65}{12}$

$\Rightarrow y = \frac{5}{12}x \pm \frac{65}{12}$

$12y = 5x \pm 65$

$5x - 12y \pm 65 = 0$

(vi) Find equation of tangent perpendicular to the line $5x - 12y = 0$

Ans. $12x + 5y \pm 65 = 0$

Sol. $m = \frac{5}{12}$, $m' = -\frac{12}{5}$

tangent $y = -\frac{12}{5}x \pm a\sqrt{1+m^2}$

$y = -\frac{12}{5}x \pm 5 \times \frac{13}{5}$

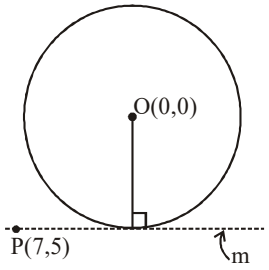
$5y = -12x \pm 65$

$12x + 5y \pm 65 = 0$

(vii) Find equation of tangent passing through (7, 5)

Ans. $m = 0, y = 5$; $m = 35/12, 35x - 12y = 185$

Sol.



$(y-5)=m(x-7)$

$y-5=mx-7m$

$mx-y+(5-7m)=0$

Apply C.O. tangency : $\left|\frac{5-7m}{\sqrt{1+m^2}}\right|=5$

$(5-7m)^2=25(1+m^2)$

$25+49m^2-70m=25+25m^2$

$24m^2-70m=0$

$2m\cdot(12m-35)=0$

$m=0$ or $m=\frac{35}{12}$

if $m=0$ tangent $y-5=0$

if $m=\frac{35}{12}$ $(y-5)=\frac{35}{12}(x-7)$

(viii) Find equation of tangent passing through (0,10)

Ans. $m=\pm\sqrt{3}$, $\sqrt{3}x-y=-10$, $\sqrt{3}x+y=10$

Sol. (Same as vii)

(ix) Find equation of tangent passing through (5,8)

Ans. $39x-80y=-445$ and $x=5$

Sol. (Same as vii)

2. Find the equation of the tangent drawn to the circle $x^2+y^2-6x+4y-3=0$ from the point (7, 4) lying outside the circle. Also find the point of contact.

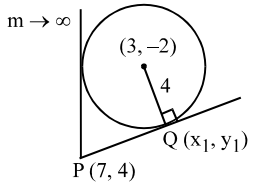
Ans. $m=\infty$, $x=7$ $(7,-2)$

$m=\frac{5}{12}$ $5x-12y=-13$; $\left(\frac{19}{13},\frac{22}{13}\right)$

Sol. $y-4=m(x-7)$

$\Rightarrow$ $mx-y-7m+4=0$

use $p=r$



$$\left|\frac{3m+2-7m+4}{\sqrt{1+m^2}}\right|=4 \quad \Rightarrow \quad \left|\frac{4m-6}{\sqrt{1+m^2}}\right|=4$$

$\Rightarrow$ $(2m-3)^2=4(1+m^2)$

$\Rightarrow$ $4m^2+9-12m=4+4m^2$

If m is finite $\Rightarrow \quad \frac{5}{12} = m$

If m is infinite then we cannot cancel $9m^2$.

$\therefore \quad m \to \infty.$

$\therefore \quad T : x = 7$

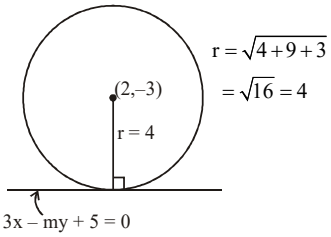
and $\quad y - 4 = \frac{5}{12}(x - 7)$

3. For what value of 'm' the line $3x - my + 5 = 0$ is tangent to the circle $x^2 + y^2 - 4x + 6y - 3 = 0$.

Ans. $m = \frac{33 \pm 4\sqrt{58}}{7}$

Sol. Apply C.O.T.

$$\left|\frac{6 + 3m + 5}{\sqrt{9 + m^2}}\right| = 4$$



$(3m + 11)^2 = 16(9 + m^2)$

$9m^2 + 121 + 66m = 144 + 16m^2$

$7m^2 - 66m + 23 = 0$

$$m = \frac{66 \pm \sqrt{66^2 - 4 \times 7 \times 23}}{2 \times 7}$$

$$m = \frac{66 \pm \sqrt{4 \times 33^2 - 4 \times 161}}{2 \times 7}$$

$$m = \frac{33 \pm \sqrt{1089 - 161}}{7} = \frac{33 \pm \sqrt{928}}{7} = \frac{33 \pm \sqrt{16 \times 58}}{7}$$

$$m = \frac{33 \pm 4\sqrt{58}}{7}$$

4. If $4l^2 - 5m^2 + 6l + 1 = 0$ then show that the line $lx + my + 1 = 0$ touches a fixed circle. Find the centre and radius of the circle.

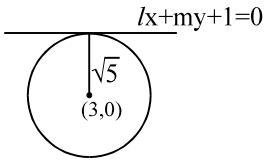
Ans. centre is (3, 0) and radius = $\sqrt{5}$

Sol. **Method-I :** $\quad 4l^2 + 6l + 1 = 5m^2$

Add $\quad 5l^2$ on both sides

$\quad 9l^2 + 6l + 1 = 5\,(l^2 + m^2)$

$\Rightarrow \quad (3l + 1)^2 = 5\,(l^2 + m^2)$



$$\therefore \quad \left|\frac{3l+1}{\sqrt{l^2+m^2}}\right| = \sqrt{5} \quad ....(1)$$

Equation (1) shows that the line lx + my + 1 touches a fixed circle whose centre is (3, 0) and radius = $\sqrt{5}$.

Method-II : Given $4l^2 - 5m^2 + 6l + 1 = 0$ ....(1)

and given line is $lx + my + 1 = 0$ ....(2)

Let line (2) touches the circle whose centre is (α, β) and radius is 'a'.

then $\frac{|l\alpha + m\beta + 1|}{\sqrt{l^2+m^2}} = a$

$\Rightarrow \quad (l\alpha + m\beta + 1)^2 = a^2(l^2 + m^2)$

$\Rightarrow \quad (\alpha^2 - a^2)\, l^2 + (\beta^2 - a)m^2 + 2lm\,\alpha\beta + 2\,\alpha\, l + 2\beta m + 1 = 0$ ....(3)

Comparing (1) and (3) we get

$\alpha^2 - a^2 = 4$

$\beta^2 - a^2 = -5$

$2\alpha = 6 \Rightarrow \quad \alpha = 3$

$2\alpha = 0 \Rightarrow \quad \beta = 0 \qquad \therefore \quad a = \sqrt{5}$

$2\alpha\beta = 0$

Hence centre of the required circle is (3, 0) and radius = $\sqrt{5}$.

5. Find the equation of the tangent to the circle $x^2 + y^2 = 4$ drawn from the point (2, 3).

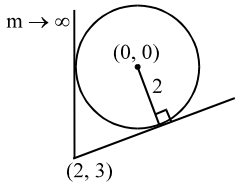
Ans. $m = \infty$, $x = 2$

$m = \frac{5}{12}$ $\quad 5x - 12y = -26$

Sol. **Method-I :** $y - 3 = m(x - 2)$

use $p = r$

$m \to \infty$; $m = 5/12$



Method-II : Equation of 'T' drawn to the circle $x^2 + y^2 = 4$ is

$y = mx \pm 2\sqrt{1+m^2}$

passes from (2, 3) ; $(3 - 2m)^2 = 4\,(1 + m)^2$

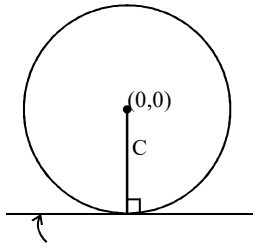
Note: With $m = 5/12$ we get 2 tangents which are parallel to each other and one of them passes through (2, 3).

6. Prove that the line $y = x + \sqrt{2}\,c$ touches the circle $x^2 + y^2 = c^2$ and find its point of contact.

Sol. Line : $y = x + \sqrt{2}c$

$x - y + \sqrt{2}c = 0$ ...(i)

Circle : $x^2 + y^2 = c^2$ ...(ii)



$y = mx + C \Rightarrow y = x + \sqrt{2}C$

Apply C.O.T.

$|p| = r$

$$\Rightarrow \left|\frac{\sqrt{2}C}{\sqrt{1+1}}\right| = C$$

$$\left|\frac{\sqrt{2}C}{\sqrt{2}}\right| = C$$

$C = C$

$\therefore$ Line (i) touches the circle (ii).

Put $y = x + \sqrt{2}C$ in (ii)

$$x^2 + \left(x + \sqrt{2}C\right)^2 = C^2$$

$$x^2 + x^2 + 2c^2 + 2\sqrt{2}cx = c^2$$

$$2x^2 + 2\sqrt{2}cx + c^2 = 0]$$

$$\left(\sqrt{2}x + c\right)^2 = 0$$

$$x = -\frac{c}{\sqrt{2}}$$

$$\text{Point of contact} \equiv \left(-\frac{c}{\sqrt{2}}, \frac{c}{\sqrt{2}}\right)$$

$$\Rightarrow y = -\frac{c}{\sqrt{2}} + \sqrt{2}c$$

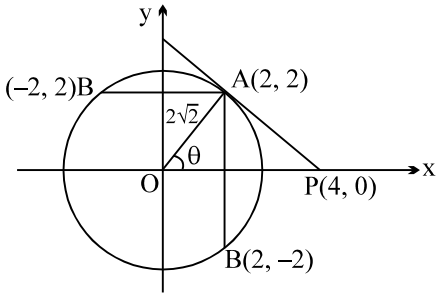
$$= \frac{-c + 2c}{\sqrt{2}} = \frac{c}{\sqrt{2}}$$

$$y = \frac{c}{\sqrt{2}}$$

7. Tangent is drawn from the point P (4, 0) to the circle $x^2 + y^2 = 8$ touches it at the point A in the 1st quadrant. Find the coordinates of another point B on the circle such that AB = 4.

Ans. $B \equiv (2, -2)$ or $(-2, 2)$

Sol. $\cos\theta = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$



$\theta = 45°$

$\therefore$ $A \equiv (2, 2)$

Let $B \equiv (x_1, y_1)$

Given $AB = 4$

$\therefore$ $(x_1 - 2)^2 + (y_1 - 2)^2 = 16$

$$x_1^2 + y_1^2 - 4x_1 - 4y_1 = 8$$

Also $x_1^2 + y_1^2 = 8$

$\therefore$ $x_1 + y_1 = 0$

$\therefore$ $2x_1^2 = 8$

$x_1 = \pm 2$

$\therefore$ $B \equiv (2, -2)$ or $(-2, 2)$

8. Find the locus of the point of intersection of the pair of tangents drawn to a circle $x^2 + y^2 = a^2$ at P (α) and Q (β), where $|\alpha - \beta| = 120°$.

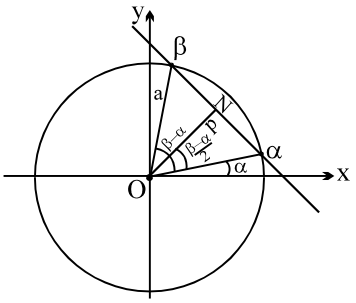
Ans. $x^2 + y^2 = 4a^2$

Sol. $h = \dfrac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$; $k = \dfrac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$ $\Rightarrow$ $h = 2a\cos\dfrac{\alpha+\beta}{2}$ & $k = 2a\sin\dfrac{\alpha+\beta}{2}$

$\therefore$ $x^2 + y^2 = 4a^2$

9. Prove that equation of a straight line joining two points α and β on the $x^2 + y^2 = a^2$ is

$$x\cos\frac{\alpha+\beta}{2} + y\sin\frac{\alpha+\beta}{2} = a\cos\frac{\alpha-\beta}{2}$$



Proof: Equation of line joining α and β is $x\cos\theta + y\sin\theta = p$

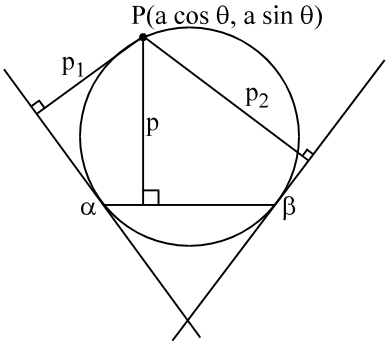
where $\theta = \alpha + \dfrac{\beta-\alpha}{2} = \dfrac{\alpha+\beta}{2}$

and $p = a\cos\dfrac{\alpha-\beta}{2}$ (from figure)

10. Prove that the perpendicular from any point on the circle $x^2 + y^2 = a^2$ on the line joining the points of contact of 2 tangents is a mean proportional between the perpendiculars from the point upon the 2 tangents. **[O]**

Sol. TPT $p^2 = p_1p_2$

Equation of chord joining α and β



$$x\cos\frac{\alpha+\beta}{2} + y\sin\frac{\alpha+\beta}{2} = a\cos\frac{\alpha-\beta}{2}$$

$$p = \left| a\left(\cos\theta\cos\frac{\alpha+\beta}{2} + \sin\theta\sin\frac{\alpha+\beta}{2} - \cos\frac{\alpha-\beta}{2}\right)\right|$$

$$p = \left| a\left(\cos\left(\frac{\alpha+\beta}{2} - \theta\right) - \cos\frac{\alpha-\beta}{2}\right)\right|$$

$$p = 2a \cdot \left| \left(\sin\left(\frac{\alpha-\theta}{2}\right) \sin\frac{\theta-\beta}{2} \right) \right| \quad ....(1)$$

Tangent at 'α' and 'β'

$x \cos\alpha + y \sin\alpha = a$ and $x \cos\beta + y \sin\beta = a$

$p_1 = |\, a(\cos\theta \cdot \cos\alpha + \sin\theta \cdot \sin\alpha - 1)\,| = a\{\cos(\theta-\alpha) - 1\}$

|||ly $p_2 = a\{\cos(\theta-\beta) - 1\}$

$\therefore$ $p_1 p_2 = a^2\{1 - \cos(\theta-\alpha)\}\{1 - \cos(\theta-\beta)\}$

$$= a^2\left(2\sin^2\frac{\theta-\alpha}{2}\right)\left(2\sin^2\frac{\theta-\beta}{2}\right)$$

$$p_1 p_2 = 4a^2 \sin^2\frac{\theta-\alpha}{2} \sin^2\frac{\theta-\beta}{2} \quad ....(2)$$

clearly $p^2 = p_1 p_2$ Hence proved

9.1 LENGTH OF TANGENT

"Length of the tangent from an external point $P(x_1, y_1)$ to a given circle:"

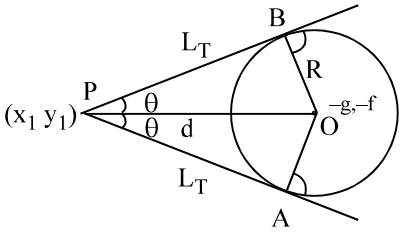
$S = x^2 + y^2 + 2gx + 2fy + c = 0$

$L^2 + R^2 = d^2$

$L^2 = d^2 - R^2 = (x_1 + g)^2 + (y_1 + f)^2 - (g^2 + f^2 - c)$

$L^2 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

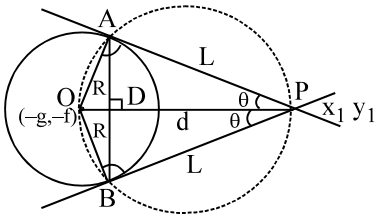
$\therefore$ $L_1 = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c} = \sqrt{S_1}$



Note: All these formulae are applicable when coeff. of x^2 & y^2 is unity

(i) Area of Quad PAOB $= 2\,\Delta$ POA

$$= 2 \cdot \frac{1}{2} R\,L = R\,L$$



(ii) Find AB i.e length of chord of contact

$$AB = 2L\sin\theta \quad \text{where } \tan\theta = \frac{R}{L}$$

$$= \frac{2RL}{\sqrt{R^2 + L^2}}$$

(iii) Area of Δ PAB (Δ formed by pair of Tangent & corresponding C.O.C.)

$$\Delta PAB = \frac{1}{2} AB \times PD = \frac{1}{2}(2L\sin\theta)(L\cos\theta) = L^2 \sin\theta\cos\theta$$

$$= \frac{RL^3}{R^2 + L^2}$$

(iv) Angle 2θ between the pair of Tangents

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2RL^2}{L(L^2-R^2)}$$

$$2\theta = \tan^{-1}\left(\frac{2RL}{L^2-R^2}\right)$$

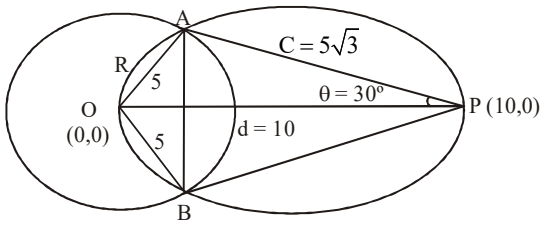
(v) Equation of the circle circumscribing the Δ PAB.
quad. PAOB is cyclic. Its circumcircle will be circumcircle of Δ PAB. For this circle, CP is diameter.
So, equation of circumcircle will be
i.e. $(x-x_1)(x+g)+(y-y_1)(y+f)=0$

Q.1 Find all these details for Circle $x^2+y^2=25$ $P(10,0)$

Ans. (i) $25\sqrt{3}$ (ii) $5\sqrt{3}$ (iii) $\frac{75\sqrt{3}}{4}$ (iv) 60° (v) $x^2+y^2-10x=0$

Sol. (i) Area of PAOB
= R.L.

$= 5\times 5\sqrt{3} = 25\sqrt{3}$



$$\tan\theta = \frac{5}{5\sqrt{3}}$$

$\theta = 30^o$

$$\sin\theta = \frac{1}{2}, \cos\theta = \frac{\sqrt{3}}{2}$$

(ii) $AB = \frac{2RL}{\sqrt{R^2+L^2}} = \frac{2\times 5\times 5\sqrt{3}}{\sqrt{25+75}}$

$= \frac{10\times 5\sqrt{3}}{10} = 5\sqrt{3}$

(iii) Area of $\Delta PAB = \frac{L^3\cdot R}{R^2+L^2} = \frac{125\times 3\sqrt{3}\times 5}{100}$

$= \frac{75}{4}\times\sqrt{3}$

(iv) Equation of circle circumscribing ΔPAB
$(x-0)(x-10)+(y-0)(y-0)=0$

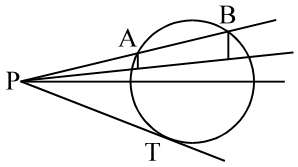
$x^2 - 10x + y^2 = 0$
$x^2 + y^2 - 10x = 0$
(v) Angle between the pairs of tangents
$2\theta = 60°$

9.2 POWER OF A POINT

Square of the length of the tangent from the point P is called power of the point P w.r.t a given circle i.e. $PT^2 = S_1$

Note: Power of a point remains constant w.r.t a circle

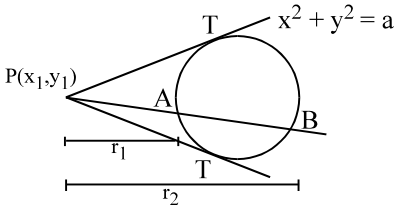
$PA \cdot PB = (PT)^2$



Analytical proof :

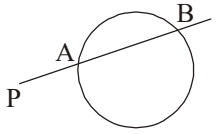
$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$; substituting $x = x_1 + r\cos\theta$ and $y = y_1 + r\sin\theta$ in $x^2 + y^2 = a^2$, we get,

$r^2 + 2r(x_1\cos\theta + y_1\sin\theta) + x_1^2 + y_1^2 - a^2 = 0$

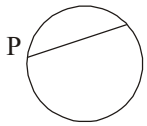


$r_1 r_2 = x_1^2 + y_1^2 - a^2 = \text{constant} = (PT)^2$

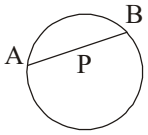
Note: Power of a point is + ve / 0 (zero) / – ve according as point 'P' lies outside / on / inside the circle.



Power > 0



Power = 0



Power < 0

Questions on Power of a point

1. Find length of tangent drawn from point P to the circle $x + y^2 + 4x + 4y + 6 = 0$
(i) P (1,1) (ii) P (0,0)

Ans. (i) 4, (ii) $\sqrt{6}$

Sol. (i) P(1, 1)

$S_1 = \sqrt{1+1+4+4+6} = 4$

(ii) P(0, 0)

$\sqrt{S_1} = \sqrt{6}$

2. Find equation of circle whose centres is (2,2) and power of origin with respect to the circle is 3.

Ans. $x^2 + y^2 - 4x - 4y + 3 = 0$

Sol. $g = -2, f = -2$

Let circle : $x^2 + y^2 - 4x - 4y + c = 0$

$\because S_1 = 3$

$x^2 + y^2 - 4x - 4y + c = 0$

$0 + 0 - 0 - 0 + c = 3$

$c = 3$

$\Rightarrow x^2 + y^2 - 4x - 4y + 3 = 0$

3. Find equation of circle such that power of (0,0), (1,0) and (0,1) to the circle are 4, 9 and 25 respectively.

Ans. $x^2 + y^2 + 4x + 20y + 4 = 0$

Sol. Let equation of circle : $x^2 + y^2 + 2gx + 2fy + c = 0$ ...(i)

P(0,0) $S_1 = 4$

$C = 4$

P(1, 0) $S_1 = 9$

$1 + 2g + c = 9$

$2g + 4 = 8$

$2g = 4$

$g = 2$

P(0,1) $S_1 = 25$

$1 + 2f + c = 25$

$2f = 25 - 5$

$f = 10$

$\therefore$ Equation of circle : $x^2 + y^2 + 4x + 20y + 4 = 0$

4. Find equation of circle such that lengths of tangent drawn from (0,0), (1,0) and (0,1) to the circle are 2, 3 and 5 respectively.

Ans $x^2 + y^2 + 4x + 20y + 4 = 0$

Sol. Let equation of circle : $x^2 + y^2 + 2gx + 2fy + c = 0$

For (0,0) $\sqrt{S_1} = 2$ for (0,1) : $\sqrt{S_1} = 5$

$S_1 = 4$ $S_1 = 25$

$C = 4$ $0 + 1 + 0 + 2f + 4 = 25$

For (1,0) $\sqrt{S_1} = 3$ $2f = 20$

$S_1 = 9$ $f = 10$

$1 + 0 + 2g + 0 + c = 9$ Equation of circle

$2g = 9 - 1 - 4$ $x^2 + y^2 + 4x + 20y + 4 = 0$

$2g = 4$

$g = 2$

5. Find the locus of a point from which length of tangents to the circles $4x^2 + 4y^2 - 9 = 0$ and $9x^2 + 9y^2 - 16 = 0$ are

in the ratio 3 : 4.

Ans. $7x^2 + 7y^2 = 20$

Sol. Let point (h, k)

$$\therefore \frac{\sqrt{4h^2+4k^2-9}}{\sqrt{9h^2+9k^2-16}} = \frac{3}{4}$$

$16(4h^2 + 4k^2 - 9) = 9(9h^2 + 9k^2 - 16)$

$64h^2 + 64k^2 = 81h^2 + 81k^2$

$x^2 + y^2 = 0$

6. Find locus of a point such that length of tangents drawn from it to the circle $x^2 + y^2 = 4$ and $x^2 + y^2 + 10x - 3y = 0$ are equal.

Ans. $10x - 3y + 4 = 0$

Sol. Let Point (h, k)

$\Rightarrow h^2 + k^2 - 4 = h^2 + k^2 + 10h - 3k$

$\Rightarrow 10x - 3y + 4 = 0$

7. Find the value of 'p' for which the power of a point P(2, 5) is negative w.r.t a circle $x^2 + y^2 - 8x - 12y + p = 0$ and the circle neither touches nor intersects the coordinate axis.

Ans $36 < p < 47$

Sol. $4 + 25 - 16 - 60 + p < 0$

$p < 47$ ....(1)

also $g^2 < c$ & $f^2 < c$

$16 < p$ ....(2)

$36 < p$ ....(3)

from (1), (2) and (3) $36 < p < 47$

8. Find the length of the Tangent from any point on the circle

$S_1 \equiv x^2 + y^2 + 2gx + 2fy + \lambda = 0$ to the circle $S_2 \equiv x^2 + y^2 + 2gx + 2fy + \mu = 0$

Ans $\sqrt{\mu - \lambda}$

Sol. From 1st circle

$r_1 = \sqrt{g^2 + f^2 - \lambda}$

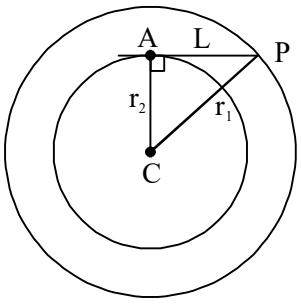
From 2nd circle

$r_2 = \sqrt{g^2 + f^2 - \mu}$

In ΔAPC

$L = \sqrt{r_1^2 - r_2^2}$

$= \sqrt{\mu - \lambda}$



9. A point moves such that a tangent from it to the circle $x^2 + y^2 + 4x - 5y + 6 = 0$ is double the length of the tangent to the circle $x^2 + y^2 = 4$. Show that the locus is a circle. Find its centre and radius.

Ans. $x^2 + y^2 - \frac{4}{3}x + \frac{5}{3}y - \frac{22}{3} = 0$; centre : $\left(\frac{2}{3}, -\frac{5}{6}\right)$; radius = $\frac{\sqrt{305}}{6}$

Sol. Let the point be $P(x_1, y_1)$ and let the lengths of the tangents from P to the two circles be t_1 and t_2 respectively.

Then $t_1^2 = x_1^2 + y_1^2 + 4x_1 - 5y_1 + 6$ and $t_2^2 = x_1^2 + y_1^2 - 4$

but $t_1 = 2t_2 \therefore$ $t_1^2 = 4t_2^2$

$\therefore$ $x_1^2 + y_1^2 + 4x_1 - 5y_1 + 6 = 4(x_1^2 + y_1^2 - 4)$

i.e. $3x_1^2 + 3y_1^2 - 4x_1 + 5y_1 - 22 = 0$

$\therefore$ locus of (x_1, y_1) is the circle, $3x^2 + 3y^2 - 4x + 5y - 22 = 0$.

Dividing throughout by 3, we get the equation as

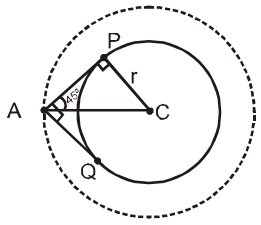
$$x^2 + y^2 - \frac{4}{3}x + \frac{5}{3}y - \frac{22}{3} = 0$$

$\therefore$ Its centre is $\left(\frac{2}{3}, -\frac{5}{6}\right)$ and radius $= \frac{\sqrt{305}}{6}$

10.1 DIRECTOR CIRCLE

The locus of the point of intersection of two perpendicular tangents is called the director circle of the given circle. The director circle of a circle is the concentric circle having radius equal to $\sqrt{2}$ times the original circle.

Proof :



$$AC = r \text{ cosec } 45^\circ = r\sqrt{2}$$

Ex. 1 Find the equation of director circle of the circle $(x - 2)^2 + (y + 1)^2 = 2$.

General Method

$y = mx + c$

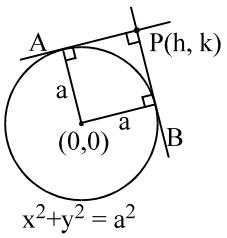
apply COT $p = r \Rightarrow$ $\left|\frac{c}{\sqrt{1+m^2}}\right| = a$

$y = mx \pm a\sqrt{1+m^2}$

passes through (h, k)

$\Rightarrow$ $(k - mh)^2 = a^2 (1 + m^2)$

$\Rightarrow$ $m^2(h^2 - a^2) - 2mhk + k^2 - a^2 = 0$



$m_1 m_2 = -1$ $\Rightarrow$ $\frac{k^2 - a^2}{h^2 - a^2} = -1$

$\Rightarrow$ $x^2 + y^2 = 2a^2$

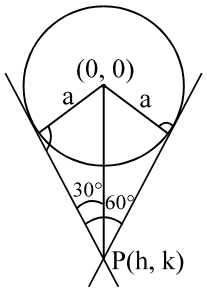
Questions on director circle

Q.1 Locus of point 'P' which moves such that the angle made by the pair of tangents drawn to the circle $x^2 + y^2 = a^2$ is 60°.

Ans. $x^2 + y^2 = 4a^2$

Sol. $\sin 30° = \dfrac{a}{\sqrt{h^2 + k^2}}$

i.e. $h^2 + k^2 = 4a^2$ or $x^2 + y^2 = 4a^2$.



Alternate : use $\tan^2 60° = \dfrac{(m_1 - m_2)^2}{(1 + m_1 m_2)^2}$

$$3 = \frac{(m_1 + m_2)^2 - 4m_1 m_2}{(1 + m_1 m_2)^2}$$

where m_1 & m_2 are the roots of the equation $(mh - k)^2 = a^2 (1 + m^2)$

Q.2 Find the range of values of 'a' such that the angles 'θ' between the pair of tangents drawn from the point (a, 0) to the circle $x^2 + y^2 = 1$ satisfies $\dfrac{\pi}{2} < \theta < \pi$.

Ans. $a \in (-\sqrt{2}, -1) \cup (1, \sqrt{2})$

Sol.

Draw the director circle of $x^2 + y^2 = 1$

i.e. $x^2 + y^2 = 2$

$\therefore$ $a \in (-\sqrt{2}, -1) \cup (1, \sqrt{2})$ Ans.

11.1 CHORD WHOSE MID POINT IS GIVEN

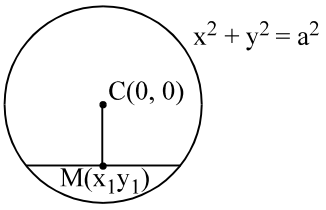
$$y - y_1 = -\frac{x_1}{y_1}(x - x_1)$$

$$y y_1 - y_1^2 = x_1^2 - x x_1$$

$$\Rightarrow \quad x x_1 + y y_1 = x_1^2 + y_1^2$$

$$\Rightarrow \quad x x_1 + y y_1 - a^2 = x_1^2 + y_1^2 - a^2$$

$$T = S_1 \qquad ....(1)$$



Note:

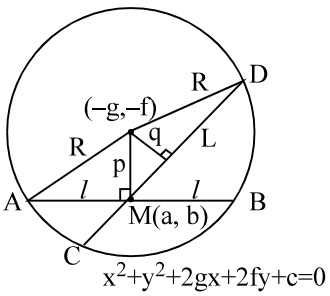
1. The shortest chord of a circle passing through a point 'M' inside the circle is one chord whose middle point is M.

2. The chord passing through a point 'M' inside the circle and which is at a maximum distance from the centre is a chord with middle point M.

3. Length of chord = $L = 2\sqrt{-S_1}$

Proof : From the figure

$$l = \sqrt{R^2 - p^2} \qquad ....(1)$$



$$L_{AB} = 2\sqrt{R^2 - p^2} = 2\sqrt{(g^2 + f^2 - c) - \{(a+g)^2 + (b+f)^2\}}$$

$$L_{AB} = 2\sqrt{\underbrace{-(a^2 + b^2 + 2ag + 2bf + c)}_{\text{negative}}}$$ (since a, b lies inside the circle)

To find mid point of a chord

Method – I

Foot of perpendicular drawn from centre on the chord is mid point of chord.

Method – II

Assume mid point P as (x_1, y_1). Write equation of chord in terms of x_1 & y_1 using $T = S_1$. Compare it with given equation & get x_1, y_1

Questions on Chord of a circle with a given middle point

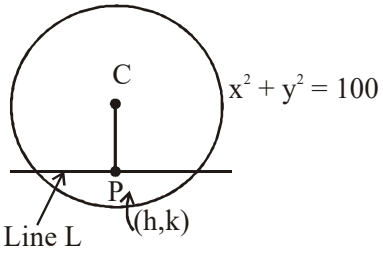
1. Find locus of mid point of a variable chord L of the circle $x^2 + y^2 = 100$ if :

(i) Chord L passes through a fixed point (a,b)

Sol. Equation of line 'L' :

$T = S_1$

$hx + ky = h^2 + k^2$ ...(i)

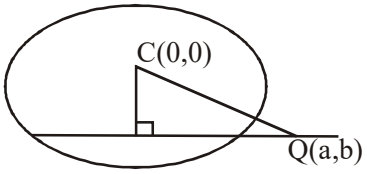


$\because$ Line 'L' passes through (a, b)

$ah + bk = h^2 + k^2$

$\Rightarrow x^2 + y^2 - ax - by = 0$

Method-II : Locus of P will be a circle for which C and D are diametrical end points.



$(x - 0)(x - a) + (y - 0)(y - b) = 0$

$x^2 + y^2 - ax - by = 0$

(ii) Slope of line L is 3

Sol. Slope of line L is 3.

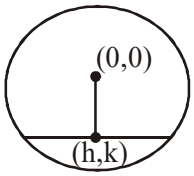
$m = 3$

$-\frac{x}{y} = 3$

$x + 3y = 0$

(iii) x-intercept of chord is 4

Sol.



(4, 0) will lie on the chord.

$T = S_1$

$hx + ky - 100 = h^2 + k^2 - 100$

$h^2 + k^2 = 4h$

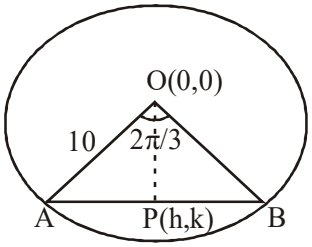
$x^2 + y^2 - 4x = 0$

(iv) chord subtends angle of $\frac{2\pi}{3}$ at center

Sol. $OP = 10 \cos 60^\circ = 5$

$h^2 + k^2 = 25$

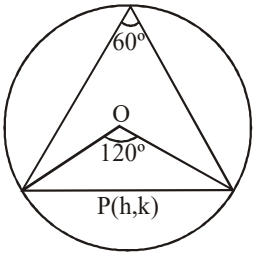
$\Rightarrow x^2 + y^2 = 25$



(v) Chord subtends angle $\frac{\pi}{3}$ at circumference.

Sol. $OP = 5$

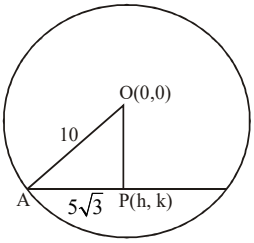
$x^2 + y^2 = 25$



(vi) Length of chord is $10\sqrt{3}$

Sol. $h^2 + k^2 = 100 - 75$

$x^2 + y^2 = 25$



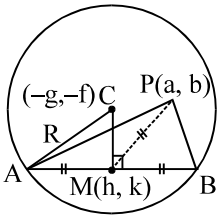
Ans. (i) $x^2+y^2 - ax - by = 0$ (ii) $x + 3y = 0$ (iii) $x^2+y^2 - 4x = 0$

(iv) $x^2+y^2=25$ (v) $x^2+y^2=25$ (vi) $x^2+y^2=25$

2 Find the equation to the locus of the middle point of the chord of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which subtends right angle at a given point P (a, b)

Ans. $(x-a)^2+(y-b)^2+x^2+y^2+2gx+2fy+c=0$

Sol. From figure



$AC^2 = AM^2 + CM^2$

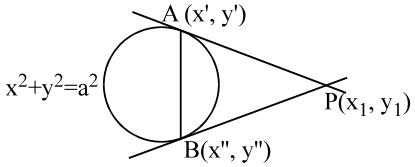
but $AM = PM$

$\therefore \quad AC^2 = PM^2 + CM^2$

$\Rightarrow \quad R^2 = (h-a)^2 + (k-b)^2 + (h+g)^2 + (k+f)^2$

11.2 CHORD OF CONTACT

Definition : It is defined as the straight line joining the point of contact of the pair of tangents drawn from an external point P (x_1, y_1) to a circle.



Note: Chord of contact will exist only if the point P does not lie inside the circle.

Equation of chord of contact AB is **T = 0**

Proof:

Tangent at 'A' $x\,x' + y\,y' = a^2$

passes through $(x_1, y_1) \Rightarrow x_1\,x' + y_1\,y' = a^2$ ....(1)

|||ly at 'B' $x\,x'' + y\,y'' = a^2$

passes through $(x_1, y_1) \Rightarrow x_1\,x'' + y_1\,y'' = a^2$ ....(2)

from (1) and (2) equation if chord of contact can be written by replacing x', x" by x and y', y" by y.

hence, $x\,x_1 + y\,y_1 = a^2$ ....(3) (same as that of tangent)

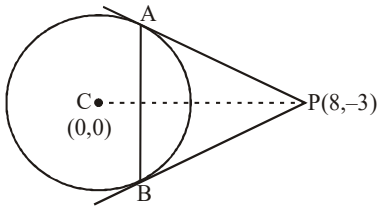
Questions

1. Tangents are drawn from the point P (8,3) to the circle $x^2 + y^2 = 49$ touching the circle at A and B. Find

(i) Equation of AB (ii) Mid point of AB

Ans. (i) $8x + 3y = 49$ (ii) (392/73, 147/73)

Sol.



Equation of AB : (Chord of contact)

$T = 0$

$\Rightarrow xx_1 + yy_1 = a^2$

$8x + 3y = 49$ ...(i)

Equation of CP : $(y-0) = \frac{3}{8}(x-0)$

$8y = 3x$ ...(ii)

P.O.I. of (i) & (ii) = Mid-point of AB = $\left(\frac{147}{73}, \frac{392}{73}\right)$

2. Tangents are drawn to the circle $x^2 + y^2 = 8$ at the points where the line $x + 2y = 4$ intersect the circle. Find the coordinates of the point of intersection of the tangents.

Ans. (2, 4)

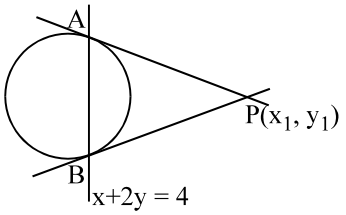
Sol. Equation of C.O.C. AB is

$x\,x_1 + y\,y_1 = 8$ ....(1)

Also equation of AB is

$x + 2y = 4$ ....(2)

(1) and (2) must be identical



$\therefore \quad \frac{x_1}{1} = \frac{y_1}{2} = \frac{8}{4} \quad \Rightarrow \quad x_1 = 2 \;;\; y_1 = 4$

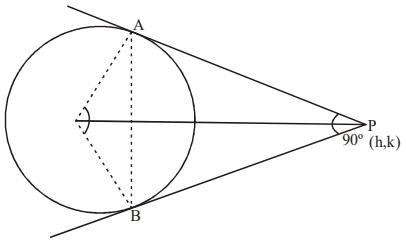
3. From on external point P, AB is chord of contact w.r.t. the circle $x^2 + y^2 = a^2$. Find locus of P

(i) if chord AB subtends right angle at centre.

Ans. $x^2 + y^2 = 2a^2$

Sol. Locus of P = Equation of director circle

$\Rightarrow x^2 + y^2 = 2a^2$



(ii) triangle PAB is equilateral triangle.

Ans. $x^2 + y^2 = 4a^2$

Sol. In ΔPOA

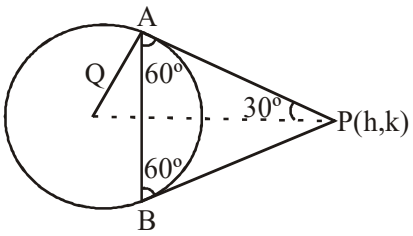
$\tan 30^\circ = \frac{OA}{PA}$

$\frac{1}{\sqrt{3}} = \frac{Q}{PA} \qquad \left(\because PA = \sqrt{S_1}\right)$

$PA^2 = 3a^2 \qquad = \sqrt{h^2 + k^2 - a^2}$

$h^2 + k^2 - a^2 = 3a^2$

$x^2 + y^2 = 4a^2$



4. Tangents are drawn to a unit circle with centre at origin from every point on the line $2x + y = 4$, prove that
(i) chord of contact passes through a fixed point
(ii) equation to the locus of the middle point of chord of contact.

Ans. (i) $\left(\frac{1}{2}, \frac{1}{4}\right)$ (ii) $2x + y = 4(x^2 + y^2)$

Sol.

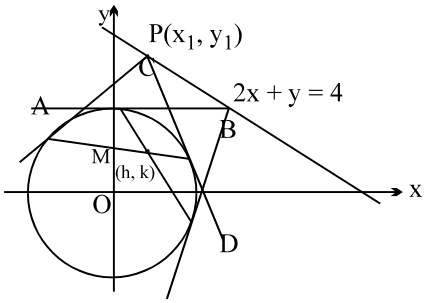
(i) 'P' is a variable point moving on line
equation of chord of contact is

$$x x_1 + y y_1 = 8 \quad ....(1)$$

also $2x_1 + y_1 = 4 \quad ....(2)$

$\therefore \quad x x_1 + y(4 - 2x_1) = 1$

$x_1(x - 2y) + (4y - 1) = 0$



$\therefore$ it represent a family of line passes through a fixed point $\left(\frac{1}{2}, \frac{1}{4}\right)$.

(ii) $M \equiv (h, k)$
equation of chord with variable given middle point (h, k)
is $xh + yk = h^2 + k^2 \quad ....(1)$
Also C.O.C is $x x_1 + y y_1 = 1 \quad ....(2)$

comparing (1) and (2) $\frac{x_1}{h} = \frac{y_1}{k} = \frac{1}{h^2 + k^2}$

$$x_1 = \frac{h}{h^2 + k^2} \; ; \; y_1 = \frac{k}{h^2 + k^2}$$

Also $2x_1 + y_1 = 4$

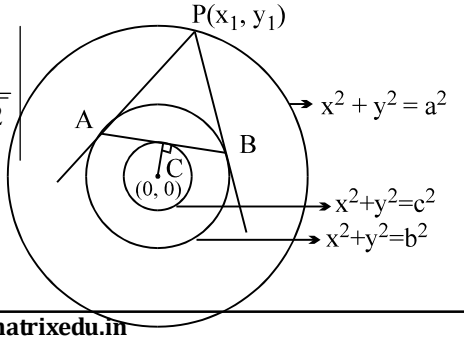
$\therefore \quad 2h + k = 4(h^2 + k^2)$

$\Rightarrow \quad 2x + y = 4(x^2 + y^2)$

5. Chord of contact of the tangent drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$. Prove that a, b, c are in G.P. (a, b, c > 0)

Sol. Equation of COC AB is $x x_1 + y y_1 = b^2$

touches the circle $x^2 + y^2 = c^2 \quad \Rightarrow c = \left| \frac{b^2}{\sqrt{x_1^2 + y_1^2}} \right|$



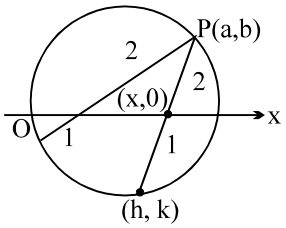
Also $x_1^2 + y_1^2 = a^2$

$\therefore \quad c = \left|\frac{b^2}{a}\right| \quad \Rightarrow b^2 = ac$ hence proved.

6. If 2 distinct chords of the circle $x^2 + y^2 - ax - by = 0$ drawn from the point (a, b) is divided by the x-axis in the ratio 2 : 1 then prove that $a^2 > 3b^2$.

Sol. We have, $0 = \frac{1\cdot b + 2k}{3} \quad \Rightarrow \quad k = -\frac{b}{2}$

hence $y = -\frac{b}{2}$ satisfy the given circle $x^2 + y^2 - ax - by = 0$



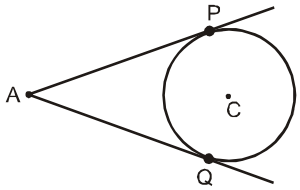
$\therefore \quad x^2 + \frac{b^2}{4} - ax + b\cdot\frac{b}{2} = 0$

$x^2 - ax + \frac{3b^2}{4} = 0$. For two distinct values of h,

$\therefore \quad D > 0 \quad \Rightarrow \quad a^2 - 3b^2 > 0 \quad \Rightarrow \quad a^2 > 3b^2$

11.3 PAIR OF TANGENTS

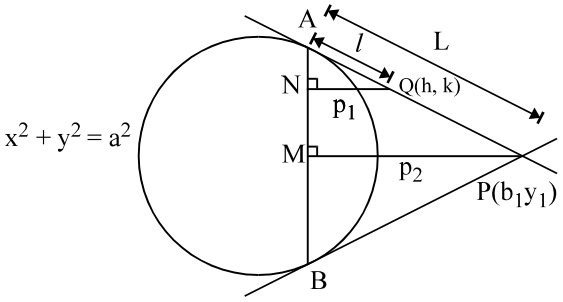
The equation of a pair of tangents drawn from the point A $(x_{1,} y_1)$ to the circle is : $\mathbf{SS_1 = T^2}$.



Proof: Equation of COC AB is $x\,x_1 + y\,y_1 = a^2$

triangles AQN and APM are similar

$$\frac{AQ}{AP} = \frac{QN}{PM} \Rightarrow \frac{l}{L} = \frac{p_1}{p_2}$$



$$\Rightarrow \frac{\sqrt{h^2 + k^2 - a^2}}{\sqrt{x_1^2 + y_1^2 - a^2}} = \frac{hx_1 + ky_1 - a^2}{x_1^2 + y_1^2 - a^2}$$

$\therefore \quad (x^2 + y^2 - a^2)\,(x_1^2 + y_1^2 - a^2) = (x\,x_1 + y\,y_1 - a^2)^2$

S $\quad$ S_1 $\quad = \quad$ T^2 $\quad$]

Q.1 Show that the equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ will be $(gx + fy)^2 = c(x^2 + y^2)$ **[O]**

Sol. $S : x^2 + y^2 + 2gx + 2fy + c$

$S_1 : x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c \Rightarrow S_1 : c$

$T : xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c \Rightarrow T : gx + fy + c$

Equation to the pair of tangent is

$SS_1 = T^2$

$(x^2 + y^2 + 2gx + 2fy + c)\cdot c = (gx + fy + c)^2$

$cx^2 + cy^2 + 2gcx + 2fcy + c^2 = g^2x^2 + f^2g^2 + c^2 + 2gfxy + 2fcy + 2gcx$

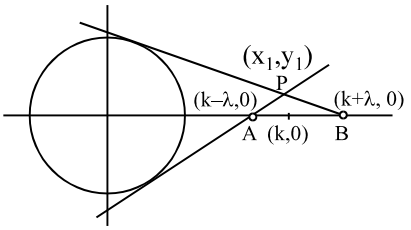
$c(x^2 + y^2) = (gx + fy)^2$

Hence, proved

Q.2 Tangents are drawn to the circle $x^2 + y^2 = a^2$ from two points on the axis of x, equidistant from the point (k, 0). Show that the locus of their intersection is $ky^2 = a^2(k - x)$ **[O]**

Sol. $SS_1 = T^2$

$(x_1^2 + y_1^2 - a^2)[(k - \lambda)^2 - a^2] = [(k - \lambda)x_1 - a^2]^2$ ....(1)



again

$(x_1^2 + y_1^2 - a^2)[(k + \lambda)^2 - a^2] = [(k + \lambda)x_1 - a^2]^2$ ....(2)

(2) – (1) gives

$4(x_1^2 + y_1^2 - a^2)\, k\lambda = [(k + \lambda)x_1 - a^2 + (k - \lambda) - a^2][(k + \lambda)x_1 - (k - \lambda)x_1]$

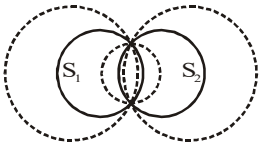
$= [2(kx_1 - a^2)][2\lambda x_1] = 4\lambda x_1 [kx_1 - a^2]$

$= k(x_1^2 + y_1^2 - a^2) = k x_1^2 - a^2x_1$

$= k y_1^2 = a^2(k - x_1)$

hence the locus is $ky^2 = a^2(k - x)$

12.1 FAMILY OF CIRCLES TYPE 1



The equation of the family of circles passing through the points of intersection of two circles $S_1 = 0$ & $S_2 = 0$ is :

$$S_1 + \lambda S_2 = 0 \qquad (\lambda \neq -1).$$

Here λ is unknown. To get λ we will be provided an extra condition

Questions on Family of Circles Type-1

1. $S_1 : x^2 + y^2 - 4x + 6y - 3 = 0$

$S_2 : x^2 + y^2 + 4x - 6y - 12 = 0$

Find equation of circle passingh through points of intersection of $S_1 = 0$ and $S_2 = 0$ and

(i) Required circle passes through origin

(ii) Centre lies on x-axis

(iii) x-coordinate of centre is 2.

(iv) line y = x is normal to the circle

(v) power of origin with respect to circle is 4.

(vi) y – coordinate of centre is 3.

(vii) radius is 5 (viii) x – intercept is 6 (ix) y - intercept is 8 (x) circle touches x-axis

(xi) circle touches y-axis

Ans. (i) $\lambda = \frac{-1}{4}$ (ii) $\lambda = 1$ (iii) $\lambda = 0$ (iv) $\lambda = 1$ (v) $\lambda = \frac{-7}{16}$ (vi) $S_2 = 0$

(vii) $g^2 + f^2 - c = 25$ (viii) $2\sqrt{g^2 - c} = 6$ (ix) $2\sqrt{f^2 - c} = 8$ (x) $g^2 = c$ (xi) $f^2 = c$

Sol. (i) Equation of circle passing through P.O.I. of $S_1 = 0$ & $S_2 = 0$

$S_1 + \lambda S_2 = 0$

$(x^2 + y^2 - 4x + 6y - 3) + \lambda(x^2 + y^2 + 4x - 6y - 12) = 0$ ...(i)

Circle passes through origin

$-3 + \lambda(-12) = 0$

$12\lambda + 3 = 0$

$$\lambda = -\frac{1}{4}$$

Equation of circle

$(1 + \lambda)x^2 + (1 + \lambda)y^2 + x(4\lambda - 4) + y\cdot(6 - 6\lambda) + (-3 - 12\lambda) = 0$

$$x^2 + y^2 + \frac{(4\lambda - 4)}{(1+\lambda)}x + \frac{(6 - 6\lambda)}{(1+\lambda)}y + \frac{(-3 - 12\lambda)}{(1+\lambda)} = 0$$

(ii) f = 0 centre lies on x-axis

$$\frac{6 - 6\lambda}{2(1+\lambda)} = 0 \Rightarrow \lambda = 1$$

(iii) x-co-ordinate of center is 2.

$\Rightarrow -g = 2$

$$-\left(\frac{2\lambda - 2}{1+\lambda}\right) = 2$$

$-2\lambda + 2 = 2 + 2\lambda \Rightarrow \lambda = 0$

(iv) Line y = x is normal to the circle

$\Rightarrow$ center lies on y = x

$\Rightarrow -g = -f$

$$\frac{2\lambda - 2}{\lambda + 1} = \frac{3 - 3\lambda}{\lambda + 1}$$

$5\lambda = 5$

$\lambda = 1$

(v) Power of origin w.r.t. circle is 4.

$S_{(0,0)} = 4$

$c = 4$

$$-\frac{(12\lambda + 3)}{\lambda + 1} = 4$$

$$\lambda = -\frac{7}{16}$$

(vi) Given :

f = 3

$$\frac{3\lambda - 3}{\lambda + 1} = 3$$

$\lambda = \infty$

$$\frac{1}{\lambda} = 0$$

$$\frac{S_1}{\lambda} + S_2 = 0$$

$\Rightarrow S_2 = 0$

(vii) radius is 5.

$g^2 + f^2 - c = 25$

$$\frac{4(\lambda-1)^2+9(1-\lambda)^2}{(\lambda+1)^2} = 25 - \frac{(3+12\lambda)}{\lambda+1}$$

$$\lambda = -\frac{9}{61}$$

(viii) x-intercept is 6.

$$2\sqrt{g^2-c} = 6$$

$g^2 - c = 9$

$g^2 = c + 9$

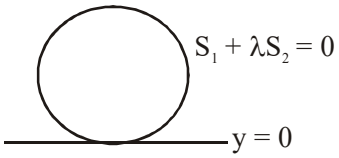
$$\left(\frac{2\lambda-2}{\lambda+1}\right)^2 = 9 - \frac{(3+12\lambda)}{\lambda+1}$$

$$\lambda = \frac{11 \pm \sqrt{177}}{14}$$

(ix) y-intercept is 8

$$2\sqrt{f^2-c} = 8$$

(x) Circle touches x-axis

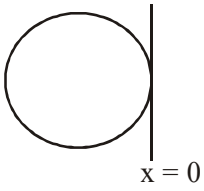


$x^2 + 2gx + c = 0$

$\Rightarrow D = 0$

$g^2 = c$

(xi) Circle touches y-axis

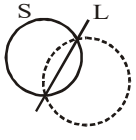


$y^2 + 2fy + c = 0$

$\Rightarrow D = 0$

$\Rightarrow f^2 = c$

12.2 FAMILY OF CIRCLES TYPE 2



The equation of the family of circles passing through the point of intersection of a circle $S = 0$ & a line $L = 0$ is given by $S + \lambda L = 0$.

Here λ is unknown. To get λ we will be provided an extra condition

Questions on Family of Circles Type-II

1. Find equation of circle passing through (3,4) and points of intersection of circle $x^2 + y^2 = 9$ and line $x + y = 3$.

Ans. $x^2 + y^2 - 9 + \lambda(x + y - 3) = 0$ & $\lambda = -4$

Sol. Equation of family of circle is

$(x^2 + y^2 - 9) + \lambda(x + y - 3) = 0$

Passing through (3, 4)

$\Rightarrow (9 + 16 - 9) + \lambda(3 + 4 - 3) = 0$

$16 + 4\lambda = 0 \Rightarrow \lambda = -4$

$\therefore$ Equation of circle :

$x^2 + y^2 - 4x - 4y + 3 = 0$

2. Find equation of circle if chord $x + y = 2$ of circle $x^2 + y^2 = 16$ is its diameter.

Ans. $x^2 + y^2 - 16 + \lambda(x + y - 2) = 0$ & $\lambda = -2$

Sol. Equation of circle :

$(x^2 + y^2 - 16) + \lambda(x + y - 2) = 0$

$x2 + y2 + \lambda x + \lambda y - 16 - 2\lambda = 0$

Centre : $\left(\frac{-\lambda}{2}, \frac{-\lambda}{2}\right)$

$\because$ Equation of diameter : $x + y = 2$

$$\frac{-\lambda}{2} - \frac{\lambda}{2} = 2$$

$-\lambda = 2$

$\lambda = -2$

$\therefore$ Equation of circle :

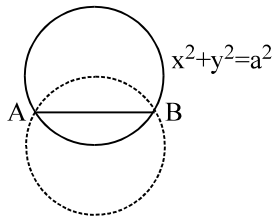
$x^2 + y^2 - 2x - 2y - 12 = 0$

3. Find the equation of a circle drawn on the chord $x \cos\alpha + y \sin\alpha = p$ of the circle $x^2 + y^2 = a^2$ as its diameter.

Ans. $x^2 + y^2 - 2p\cos\alpha\, x - 2p\sin\alpha\, y + 2p^2 - a^2 = 0$

Sol. Equation of family of circles is $(x^2 + y^2 - a^2) + \lambda(x\cos\alpha + y\sin\alpha - p) = 0$

Now centre lies on the line $x\cos\alpha + y\sin\alpha = p$



i.e. $-\frac{\lambda}{2}\cos^2\alpha - \frac{\lambda}{2}\sin^2\alpha = p$

$\Rightarrow \quad \lambda = -2p$

4. Show that the equation $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$ represents for different values of λ, a system of circles passing through two fixed points A and B on the x-axis, and also find the equation of that circle of the system the tangent to which at A and B meet on the line $x + 2y + 5 = 0$. **[O]**

Ans A ≡ (4, 0) B ≡ (–2, 0); $x^2 + y^2 - 2x - 6y - 8 = 0$

Sol. $(x^2 + y^2 - 2x - 8) - 2\lambda \cdot y = 0$

$S + \lambda L = 0$

Solving $S = 0$ & $L = 0$

put $y = 0 \Rightarrow x^2 - 2x - 8 = 0$

$(x - 4)(x + 2) = 0$

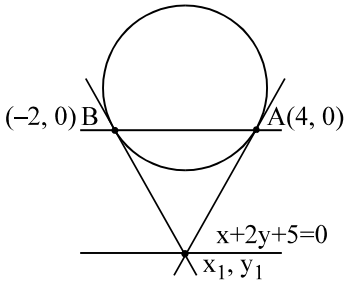
$x = -2$ or 4

$\therefore$ A ≡ (4, 0)

B ≡ (–2, 0)

Equation of AB :

C.O.C. $xx_1 + yy_1 - 1(x + x_1) - \lambda(y + y_1) - 8 = 0$



$\Rightarrow \quad x(x_1 - 1) + y(y - y_1) - (x_1 + \lambda y_1 + 8) = 0 \quad ....(1)$

Also equation of AB is x-axis i.e

$0 \cdot x + 1 \cdot y + 0 \cdot c = 0 \quad ....(2)$

comparing (1) & (2)

$$\frac{0}{x_1 - 1} = \frac{1}{y_1 - \lambda} = \frac{0}{x_1 + \lambda y_1 + 8}$$

$\therefore \quad x_1 = 1 \ \& \ \lambda y_1 + 9 = 0$

$$y_1 = \frac{-9}{\lambda}$$

Also $x_1 + 2y_1 + 5 = 0 \Rightarrow 1 - \frac{18}{\lambda} + 5 = 0 \quad \Rightarrow \quad \lambda = \frac{18}{\lambda} \quad \Rightarrow \quad \lambda = 3$

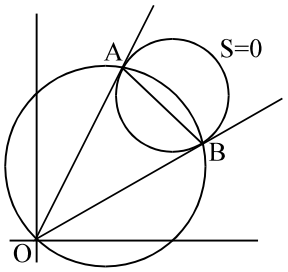
$\therefore$ equation of circle is $x^2 + y^2 - 2x - 6y - 8 = 0$

5. Find the equation of a circle which passes through origin and through the point of contact of the tangents drawn from the origin to the circle $x^2 + y^2 - 11x + 13y + 17 = 0$

Ans $x^2 + y^2 - 11x + 13y + 17 + \lambda(11x - 13y - 34) = 0; \quad \lambda = 1/2$

$2x^2 + 2y^2 - 11x + 13y = 0$

Sol. Equation of COC AB is



$$x \cdot 0 + y \cdot 0 - \frac{11}{2}(x + 0) + \frac{13}{2}(y + 0) + 17 = 0$$

i.e. $11x - 13y - 34 = 0$

Now equation of family of circle is

$x^2 + y^2 - 11x + 13y + 17 + \lambda(11x - 13y - 34) = 0$

passes through (0, 0)

$\lambda = 1/2$]

Note : Converse of type – 2

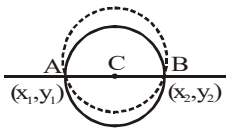
A variable circle $S + \lambda L = 0$ will always pass through points of intersection of circle $S = 0$ & line $L = 0$.

Note : Converse of type – 2

A variable circle $S + \lambda L = 0$ will always pass through points of intersection of circle $S = 0$ & line $L = 0$.

12.3 FAMILY OF CIRCLES TYPE 3

The equation of a family of circles passing through two given points (x_1, y_1) & (x_2, y_2) can be written in the form :

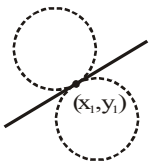


$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + K \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$ where K is a parameter.

12.4 FAMILY OF CIRCLES TYPE 4

Equation of family of circles touching a line at its fixed point (x_1, y_1) is:

$(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$

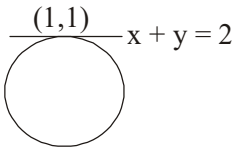


Questions on Type-4

1. Find equation of circle passing through origin and touching the line $x + y = 2$ at (1, 1)

Ans. $x^2 + y^2 - x - y = 0$, $\lambda = 1$

Sol.



Equation of circle

$(x - 1)^2 + (y - 1)^2 + l(x + y - 2) = 0$ ...(i)

$\because$ Circle passing through (0, 0)

$\therefore\ 2 + \lambda(-2) = 0$

$\lambda = 1$

$\Rightarrow x^2 + y^2 - x - y = 0$

2. Find equation of circle touching the line $y = 3x$ at origin and centre lies on the line $x + y = 2$.

Ans. $x^2 + y^2 - 6x + 2y = 0$, $\lambda = -2$

Sol. Equation of circle is

$\Rightarrow (x - 0)^2 + (y - 0)^2 + \lambda(y - 3x) = 0$

$x^2 + y^2 - 3\lambda x + \lambda y = 0$

$\because$ Center lies on $x + y = 2$

$\Rightarrow \left(\frac{3\lambda}{2}, -\frac{\lambda}{2}\right)$ lies on $x + y = 2$

$\Rightarrow \frac{3\lambda}{2} - \frac{\lambda}{2} = 2$

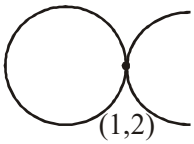
$\lambda = 2$

Equation of circle : $x^2 + y^2 - 6x + 2y = 0$

3. Find equation of circle passing through (2,0) and touching the parabola $y^2 = 4x$ at (1,2).

Ans. $(x-1)^2 + (y-2)^2 + \lambda(x - y + 1) = 0$; $\lambda = \frac{-5}{3}$

Sol.



For $y^2 = 4x$

$T = 0$ at (1, 2)

$2(x + x_1) = y \cdot y_1$

$2(x + 1) = 2y$

$x - y + 1 = 0$

Equation of circle

$(x - 1)^2 + (y - 2)^2 + \lambda(x - y + 1) = 0$ ...(i)

Circle passing through (2, 0)

$\therefore \lambda = -\frac{5}{3}$

Equation of circle : $(x-1)^2 + (y-2)^2 - \frac{5}{3}(x-y+1) = 0$

4. Find the equation of a circle which touches the line $2x-y=4$ at the point $(1,-2)$ and
(a) Passes through (3, 4) (b) Radius = 5

Ans. (a) $x^2 + y^2 + 38x - 16y - 75 = 0$

(b) $\lambda = 2\sqrt{5} \Rightarrow x^2 + y^2 + x\left(4\sqrt{5}-2\right) + y\left(4-2\sqrt{5}\right) + 5 - 8\sqrt{5} = 0$

$\lambda = -2\sqrt{5} \Rightarrow x^2 + y^2 - x\left(4\sqrt{5}+2\right) + y\left(4+2\sqrt{5}\right) + 5 + 8\sqrt{5} = 0$

Sol. $(x-1)^2 + (y+2)^2 + \lambda(2x-y-4) = 0$ ...(i)

(a) Passing through (3, 4), get λ

(b) radius = $5 = \sqrt{g^2 + f^2 - c}$

5. Find the equations of the circle which passes through the point $(-1, 2)$ & touches the circle $x^2 + y^2 - 8x + 6y = 0$ at origin.

Ans $(x-0)^2 + (y-0)^2 + \lambda(4x-3y) = 0$; $\lambda = 1/2$

Sol. Equations of tangent is: $4x - 3y = 0$

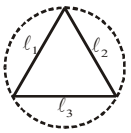
$\therefore$ equation of the family of the circle is

$(x-0)^2 + (y-0)^2 + \lambda(4x-3y) = 0$

passes through P $(-1, 2)$ & get $\lambda = 1/2$

Note

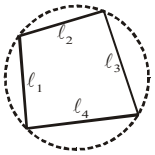
1. Family of circles circumscribing a triangle whose sides are given by



$L_1 = 0$; $L_2 = 0$ & $L_3 = 0$ is given by ; $L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$

provided coefficient of $xy = 0$ & coefficient of x^2 = coefficient of y^2.

2. Equation of circle circumscribing a quadrilateral whose sides in order are



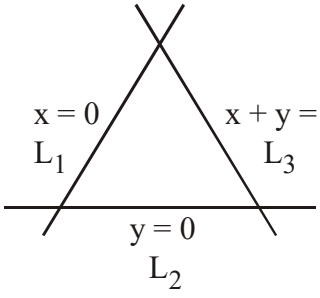
represented by the lines $L_1 = 0, L_2 = 0, L_3 = 0$ & $L_4 = 0$ is $L_1L_3 + \lambda L_2L_4 = 0$

provided coefficient of x^2 = coefficient of y^2 and coefficient of $xy = 0$.

Q.6 Find equation of circle circumscribing triangle formed by lines $x = 0$, $y = 0$ and $x + y = 4$

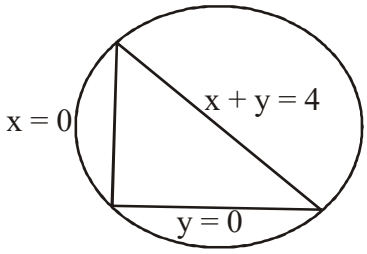
Ans. $x^2 + y^2 - 4x - 4y = 0$

Sol



$\mu = \lambda = -1/2$

Equation of circum circle is



$x \cdot y + \lambda \cdot y \cdot (x + y - 4) + \mu x \cdot (x + y - 4) = 0$

$xy + \lambda xy + \lambda y^2 - 4\lambda y + \mu x^2 + 4\mu y - 4\mu x = 0$

$\mu x^2 + \lambda y^2 + (11\lambda + \mu)xy - 4\mu x - 4\lambda y = 0$

$\mu = \lambda,\ 1 + \lambda + \mu = 0$

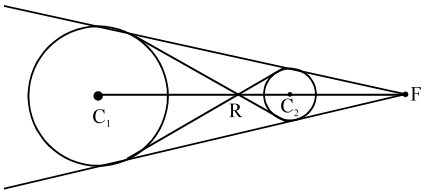
$2\lambda = -1$

$$\lambda = -\frac{1}{2} = \mu$$

$$(\therefore \text{ Equation : } -\frac{x^2}{2} - \frac{y^2}{2} + 2x + 2y = 0$$

$x^2 + y^2 - 4x - 4y = 0$

13.1 COMMON TANGENTS



There are two types of common tangents

1. Direct common tangent (DCT)

L_1 & L_2 are DCTs. These are also called external common tangent. Both centres lie on same side of DCTs. Point of intersection of DCTs (P) divides C_1C_2 in $r_1 : r_2$ externally.

If $r_1 = r_2$, then DCT will be parallel to line joining centers.

2. Transverse common tangent (TCT)

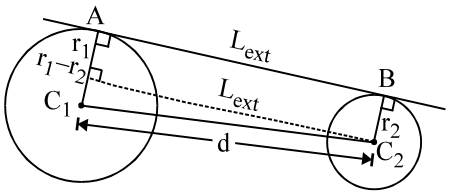
L_3 & L_4 are TCT. These are also called internal commom tangents centres lie on opposite side of a TCT

Point of intersection of TCTs (i.e.R) divides C_1C_2 in $r_1 : r_2$ internally.

Length of common tangents

To find analytically the length of common tangent (both external and internal common tangent)

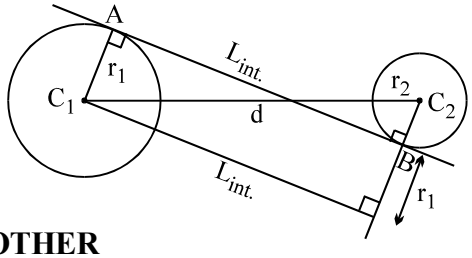
From figure : $L_{ext}^2 + (r_1 - r_2)^2 = d^2$



$$\Rightarrow \quad L_{ext} = \sqrt{d^2 - (r_1 - r_2)^2}$$

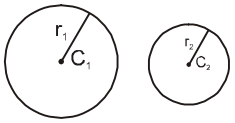
$\therefore \quad d^2 = L_{int}^2 + (r_1 + r_2)^2$

$\therefore \quad L_{int} = \sqrt{d^2 - (r_1 + r_2)^2}$



13.2 POSITION OF TWO CIRCLES RELATIVE TO EACH OTHER

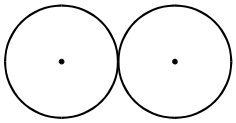
1. If circles are separated from each other.



4 common tangents (2 direct and 2 transverse)

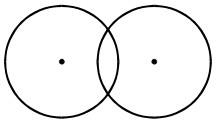
$r_1 + r_2 < c_1\, c_2$.

2. If circles touch each other externally.



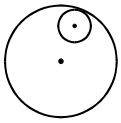
3 common tangents. $r_1 + r_2 = c_1\, c_2$.

3. If circles cut each other at 2 common points



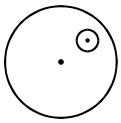
2 common tangents. $|r_1 - r_2| < c_1\, c_2 < r_1 + r_2$

4. If one circle touch another circle internally



1 common tangent. $|r_1 - r_2| = c_1\, c_2$.

5. If one circle is contained in another circle.



No common tangent. $c_1\, c_2 < |r_1 - r_2|$.

Questions on common tangents

1. $C_1 : x^2 + y^2 = 9$
   $C_2 : x^2 + y^2 - 8x + 7 = 0$
   $C_3 : x^2 + y^2 - 10y + 24 = 0$
   Find number of common tangents and length of common tangent.

(i) C_1 & C_2

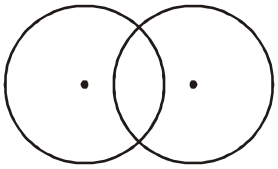
(ii) C_2 & C_3

(iii) Find equation of DCT of C_1 & C_3

Ans. (i) No of common tangents = 2 and L = 4 (ii) No of common tangents = 4 and $L_{DCT} = \sqrt{37}$ and $L_{TCT} = 5$

(iii) $2y = \pm\sqrt{21}x + 15$

Sol. (i) $C_1 : (0, 0); r_1 = 3$
$C_2 : (4, 0); r_2 = 3$
$d = C_1C_2 = 4$, $r_1 + r_2 = 6$, $|r_1 - r_2| = 0$
$\because\ 0 < d < r_1 + r_2$
$|r_1 - r_2| < C_1C_2 < r_1 + r_2$



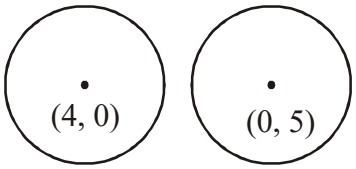
No. of common tangent = 2
2 DCT and 0 TCT

$L_{ext} = \sqrt{d^2 - (r_1 - r_2)^2} = 4$

(ii) C_2 (4, 0), $r_2 = 3$
C_3 (0, 5), $r_3 = 1$

$d = C_1C_2 = \sqrt{16+25} = \sqrt{41}$

$r_2 + r_3 = 4$; $|r_2 - r_3| = 2$
$\Rightarrow d > r_2 + r_3$

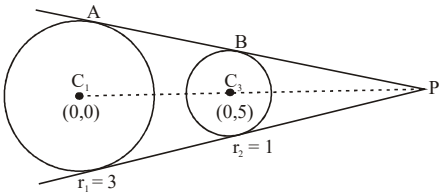


4 Common Tangents
2 DCT
2 TCT

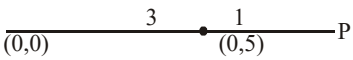
$L_{ext.} = \sqrt{d^2 - (r_2 - r_3)^2} = \sqrt{41-4} = \sqrt{37}$

$L_{int.} = \sqrt{d^2 - (r_2 + r_3)^2} = \sqrt{41-16} = 5$

(iii)



P divides C_1C_3 in ratio $r_1 : r_3$ externally



Co-ordinate of P : $P\left(\frac{3\cdot0 - 0\cdot1}{3-1}, \frac{3\cdot5 - 1\cdot0}{3-1}\right)$

$= \left(0, \frac{15}{2}\right)$

Equation is

$y - \frac{15}{2} = m(x - 0)$

Apply C.O.T.

$$\left|\frac{\frac{15}{2}}{\sqrt{1+m^2}}\right| = 3$$

$$5 = 2\sqrt{1+m^2}$$

$$\Rightarrow \frac{25}{4} - 1 = m^2$$

$$m = \pm\frac{\sqrt{21}}{2}$$

Equation of circle is

$$y = \pm\frac{\sqrt{21}x}{2} + \frac{15}{2}$$

2. Find the range of 'r' so that the circles : $(x-1)^2 + (y-3)^2 = r^2$ and $(x-4)^2 + (y+1)^2 = 9$ intersects at 2 distinct points ('r' being the radius of the circle.)

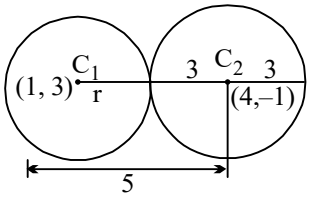
Ans $2 < r < 8$

Sol. **Method-I :** for circles to intersecting

$r > 5 - 3$

and $r < 8$

$\therefore$ range of 'r' $2 < r < 8$



Method-II : $d = 5$

$\therefore$ for circle to intersect

$|r_1 - r_2| < d < r_1 + r_2$

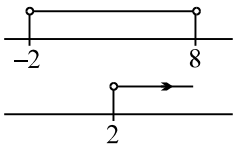
$\Rightarrow$ $|r - 3| < 5 < r + 3$

$\therefore$ $r > 2$ ....(1)

and $|r - 3| < 5 \Rightarrow -5 < (r-3) < 5$

$-2 < r < 8$ ....(2)

$\therefore$ $2 < r < 8$

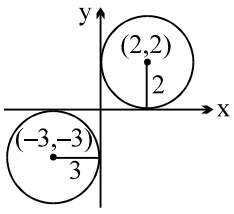


3. Find equation of transverse common tangent of circles $x^2 + y^2 - 4x - 4y + 4 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$

Ans $x = 0$ and $y = 0$

Sol

from figure



4. Prove that the common tangents to the circles $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 + 2x = 0$ from an equilateral triangle.

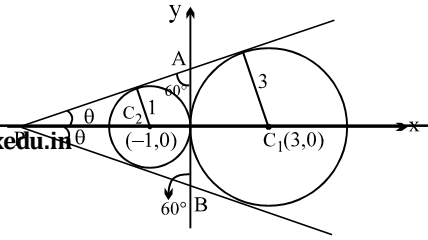
Sol. $S_1 : x^2 + y^2 - 6x = 0$ $C_1 \equiv (3, 0)$; $r_1 = 3$

$S_2 : x^2 + y^2 + 2x = 0$ $C_2 \equiv (-1, 0)$; $r_2 = 1$

$d = 4$

$\therefore$ $d = r_1 + r_2$

$\Rightarrow$ circles touches externally



$$\left.\begin{array}{l}\sin\theta = \dfrac{1}{PC_2} \\ \text{and} \quad \sin\theta = \dfrac{3}{PC_1}\end{array}\right\} \Rightarrow \quad PC_1 - PC_2 = 2\,\text{cosec}\theta$$

$\Rightarrow$ $C_1 C_2 = 2\,\text{cosec}\theta$

$\Rightarrow$ $4 = 2\,\text{cosec}\theta$

$\Rightarrow$ $\sin\theta = \dfrac{1}{2} \quad \Rightarrow \quad \theta = 3\theta \Rightarrow \quad 2\theta = 60°$

$\therefore$ ABP is equilateral.

5. Find all the common tangent to the circles $x^2 + y^2 = 1$ and $(x-1)^2 + (y-3)^2 = 4$. **[O]**

Ans. $y = 1$

$3x + 4y = 5$

$x + 1 = 0$

$4x - 3y = 5$

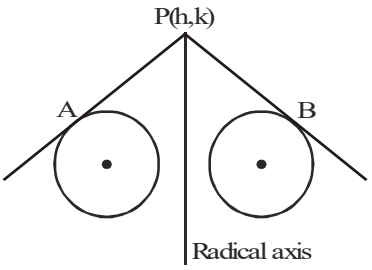
6. Find the equation of the circles to which the line $4x + 3y = 10$ is a common tangent at $(1, 2)$ and radius of each of the circle is 5. **[O]**

$x^2 + y^2 - 10x - 10y + 25 = 0$

$x^2 + y^2 + 6x + 2y - 15 = 0$

14.1 RADICAL AXIS

The locus of a point, which moves in such a way that its power w.r.t. to the two circles are equal, is called the radical axis.



$S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1 y + c_1 = 0$

$S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$

Let P(h,k) is a point and PA,PB are length of two tangents on the circles from point P, Then from definition -

$\sqrt{h^2 + k^2 + 2g_1h + 2f_1k + c_1} = \sqrt{h^2 + k^2 + 2g_2h + 2f_2k + c_2}$ or $2(g_1 - g_2)\,h + 2(f_1 - f_2)\,k + c_1 - c_2 = 0$

$\therefore$ locus of (h,k)

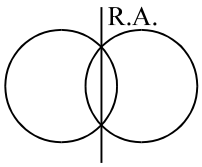
$2x(g_1 - g_2) + 2y(f_1 - f_2)k + c_1 - c_2 = 0$

$\mathbf{S_1 - S_2 = 0}$ which is the equation of radical axis.

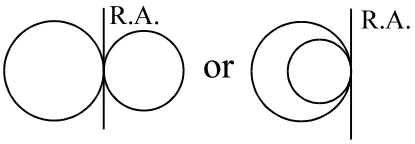
Important Points

1. If two circles intersect, then the radical axis is the common

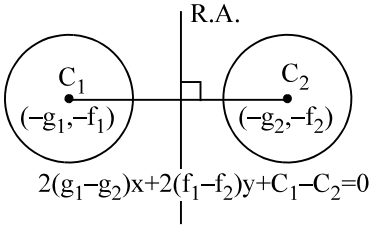
chord of the two circles.



2. If two circles touch each other then the radical axis is the common tangent of the two circles at the common point of contact.



3. Radical axis is always perpendicular to the line joining the centres of the two circles.



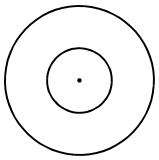
4. Radical axis need not always pass through the mid point of the line joining the centres of the two circles.

5. Radical axis bisects a common tangent between the two circles.

6. Pairs of circle which do not have radical axis are concentric.

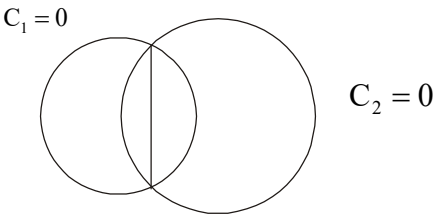
$S_1 = x^2 + y^2 + 2gx + 2fy + c_1 = 0$

$S_2 = x^2 + y^2 + 2gx + 2fy + c_2 = 0$



7. If one circle is contained in another circle when radical axis passes outside to both the circles.

8. If circle $C_2 = 0$ bisects the circumference of circle $C_1 = 0$, then centre of $C_1 = 0$ will lie on their common chord

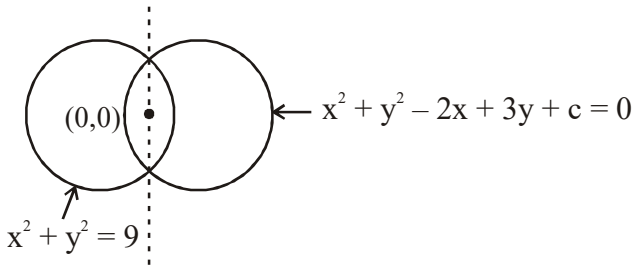


Questions on Radical Axis

1. If circle $x^2 + y^2 - 2x + 3y + c = 0$ bisects the circumference of circle $x^2 + y^2 = 9$, find c.

Ans $c = -9$

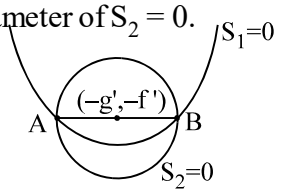
Sol.



$S_1 - S_2 = 0$
$x^2 + y^2 - 9 - x^2 - y^2 + 2x - 3y - c = 0$
$2x - 3y - 9 - c = 0$
(0, 0) will lie on R.A.
$\Rightarrow c = -9$

2. Prove that the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ will bisect the circumference of the circle $x^2 + y^2 + 2g'x + 2f'y + c' = 0$ if $2g'(g - g') + 2f'(f - f') = c - c'$.

Sol. $S_1 : x^2 + y^2 + 2gx + 2fy + c = 0$
$S_2 : x^2 + y^2 + 2g'x + 2f'y + c' = 0$
If S_1 bisects the circumference of S_2 then their common chord is the diameter of $S_2 = 0$.



equation of common chord
$S_1 - S_2 = 0$
$2(g - g')x + 2(f - f')y + c - c' = 0$ ....(1)
(–g, – f') lies on equation (1)
$-2g'(g - g') - 2f'(f - f') + c - c' = 0$
$\therefore$ $2g'(g - g') + 2f'(f - f') = c - c'$ hence proved.

3. Find the equation of a circle which bisects the circumferences of the circles $x^2 + y^2 = 1$, $x^2 + y^2 + 2x = 3$ and $x^2 + y^2 + 2y = 3$.

Ans. $x^2 + y^2 + 4x + 4y - 1 = 0$

Sol. Let equation of circle is
$x^2 + y^2 + 2gx + 2fy + c = 0$
(i) for $x^2 + y^2 = 1$
R.A. $2gx + 2fy + c + 1 = 0$
(0, 0) will lie on R.A.
$\Rightarrow c + 1 = 0$
$c = -1$
(ii) $x^2 + y^2 + 2x - 3 = 0$
Equation of R.A.
$S_1 - S_2 = 0$
$(2g - 2)x + 2fy + c + 3 = 0$
(–1, 0) lie on R.A.
$\Rightarrow 2 - 2g + 2 = 0$
$\Rightarrow 4 = 2g$
$g = 2$
(iii) $x^2 + y^2 + 2y = 3$
equation of R.A.
$2gx + (2f - 2)y + c + 3 = 0$
(0, –1) lie on R.A.
$f = 2$

$\Rightarrow$ Equation of circle

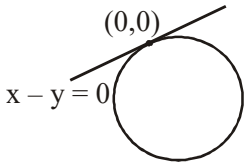
$x^2 + y^2 + 4x + 4y - 1 = 0$

4. Find the equation of the circle which bisects the circumference of the circle $x^2 + y^2 + 2y - 3 = 0$ and touches the line $x - y = 0$ at origin

Ans. $x^2+y^2 + \lambda (x - y) = 0$; $\lambda = -5$.

$x^2 + y^2 - 5 (x - y) = 0$

Sol.



Equation of R.A.

$x^2 + y^2 + \lambda(x - y) - x^2 - y^2 - 2y + 3 = 0$

$\lambda x - \lambda y - 2y + 3 = 0$

$\lambda x - y(\lambda + 2) + 3 = 0$

$(0, -1)$ will lie on R.A.

$\Rightarrow \lambda + 2 + 3 = 0$

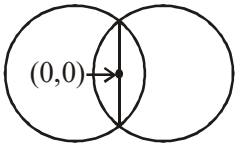
$\lambda = -5$

Equation of circle is $x^2 + y^2 - 5x + 5y = 0$

5. Find the locus of the centre of circles which bisect the circumference of the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 2x + 6y + 1 = 0$.

Ans. $2x - 6y - 15 = 0$

Sol.



Let centre : (g, f)

$S_1 : x^2 + y^2 - 2gx - 2fy + c = 0$

$S_2 : x^2 + y^2 - 4 = 0$

R-axis

$-2gx - 2fy + (c + 4) = 0$

$(0, 0)$ will satisfy

$c + 4 = 0$

$c = -4$

$S_3 : x^2 + y^2 - 2x + 6y + 1 = 0$

R-axis

$-2gx + 2x - 2fy - 6y + c - 1 = 0$

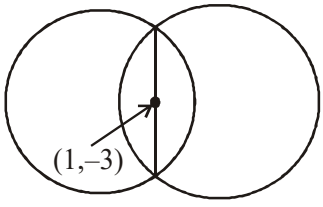
$(1, -3)$ will lie on R-axis.

$-2g + 2 + 6f + 18 - 5 = 0$

$2g - 6f = 15$

Locus of center is

$2x - 6y - 15 = 0$



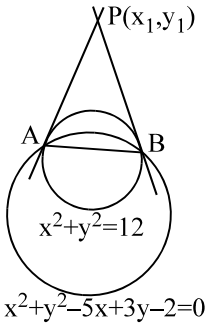
6. Tangent are drawn to the circle $x^2 + y^2 = 12$ at the points where it is met by the circle $x^2 + y^2 - 5x + 3y - 2 = 0$. Find the point of intersection of the tangents.

Ans $\left(6, -\frac{18}{5}\right)$

Sol. equation of AB :

$(x^2 + y^2 - 5x + 3y - 2) - (x^2 + y^2 - 12) = 0$

$5x - 3y - 10 = 0$ ....(1)



Also AB is chord of contact w.r.t the circle $x^2 + y^2 = 12$

i.e. $x x_1 + y y_1 - 12 = 0$ ....(2)

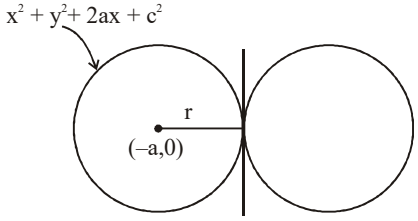
comparing (1) and (2)

$$\frac{x_1}{5} = \frac{y_1}{-3} = \frac{12}{10}$$

$x_1 = 6$ & $y_1 = -\frac{18}{5}$

7. Prove that the circle $x^2 + y^2 + 2ax + c^2 = 0$ and $x^2 + y^2 + 2by + c^2 = 0$ touche each other if $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$

Sol. Subtract to get common tangent and drop perpendicular from centre on any one circle and equate it to its radius.



Equation of R.A.

$S_1 - S_2 = 0$

$x^2 + y^2 + 2ax + c^2 - x^2 - y^2 - 2by - c^2 = 0$

$ax - by = 0$

Apply C.O.T. $= \left|\frac{-a^2}{\sqrt{a^2 + b^2}}\right| = \sqrt{a^2 + 0 - c^2}$

$$\frac{a^4}{a^2 + b^2} = a^2 - c^2$$

$a^4 = (a^2 - c^2)(a^2 + b^2)$

$a^4 = a^4 + a^2b^2 - c^2a^2 - b^2c^2$

$a^2b^2 = c^2a^2 + b^2c^2$

$\div$ by $a^2b^2c^2$

$$\frac{1}{c^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

8. Consider a family of circles passing through two fixed points A (3, 7) and B (6, 5). Show that the chords in which the circles $x^2 + y^2 - 4x - 6y - 3 = 0$ cuts the member of the family are concurrent at a point. Also find the coordinates of this point.

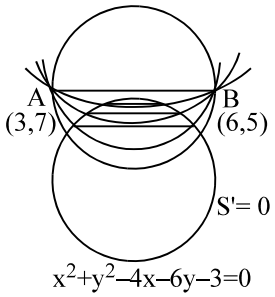
Ans. $\left(2, \frac{23}{3}\right)$

Sol. Family of circles passes through two fixed point is

$S + \lambda L = 0$ ....(1)

where $S = (x-3)(x-6) + (y-7)(y-5) = 0$

$$L = \begin{vmatrix} x & y & 1 \\ 3 & 7 & 1 \\ 6 & 5 & 1 \end{vmatrix} = 0$$



Equation of variable common chord is

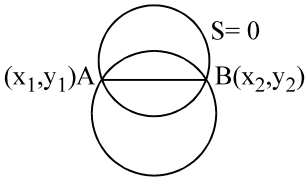
$S + \lambda L - S' = 0$

$\Rightarrow$ $(S - S') + \lambda L = 0$

or $L' + \lambda L = 0$ which represents family of lines concurrent at $L' = 0$ and $L = 0$.

9. Given $S = x^2 + y^2 + 2gx + 2fy + c = 0$ and two point A and B on it. Prove that the equation of the line AB is $(x_1 + x_2 + 2g)x + (y_1 + y_2 + 2f)y = x_1x_2 + y_1 y_2 - c$ **[O]**

Sol

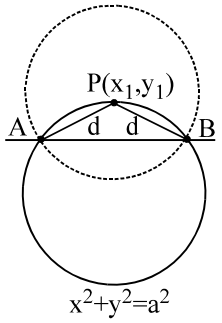


10. Show that the equation of a straight line meeting the circle $x^2 + y^2 = a^2$ in 2 points at equal distance 'd' from the point (x_1, y_1) on its circumference is : $x\,x_1 + y\,y_1 - a^2 + \frac{d^2}{2} = 0$ **[O]**

Sol. Equation of a circle with centre 'p' and radius 'd'

i.e. $(x - x_1)^2 + (y - y_1)^2 = d^2$

or $x^2 + y^2 - 2xx_1 - 2yy_1 + x_1^2 + y_1^2 = d^2$; also $x_1^2 + y_1^2 = a^2$



the required line is the radical axis between the circles

$x^2 + y^2 - 2xx_1 - 2yy_1 + a^2 - d^2 = 0$ ....(1)

and $x^2 + y^2 - a^2 = 0$ ....(2)

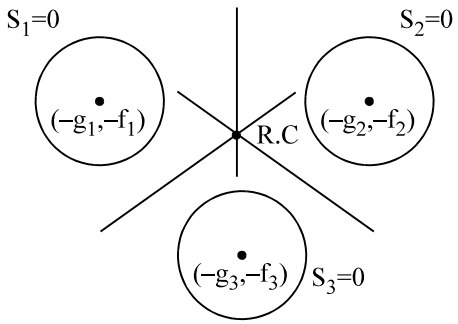
$\therefore$ equation of AB is (1) – (2)

$-2\,x\,x_1 - 2\,y\,y_1 + 2a^2 - d^2 = 0$

i.e. $x\,x_1 + y\,y_1 - a^2 + \frac{d^2}{2} = 0$ Hence proved

14.2 RADICAL CENTER

The common point of intersection of the radical axis of 3 circles taken 2 at a time is called the ***Radical Centre*** of three circles.



Radical Axis of S_1 & S_2 : $2(g_1 - g_2)x + 2(f_1 - f_2)y + c_1 - c_2 = 0$

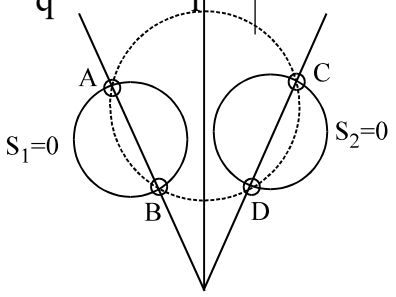
Radical Axis of S_2 & S_3 : $2(g_2 - g_3)x + 2(f_2 - f_3)y + c_2 - c_3 = 0$
Radical Axis of S_3 & S_1 : $2(g_3 - g_1)x + 2(f_3 - f_1)y + c_3 - c_1 = 0$

$$D = \begin{vmatrix} g_1 - g_2 & f_1 - f_2 & c_1 - c_2 \\ g_2 - g_3 & f_2 - f_3 & c_2 - c_3 \\ g_3 - g_1 & f_3 - f_1 & c_3 - c_1 \end{vmatrix} = 0 \quad ; \text{ Use } R_1 \to R_1 + R_2 + R_3 \quad \Rightarrow \quad D = 0$$

(solve any 2 radical axes to get radical centre which is a point from which tangent to the three circles are equal.)

1. Lines $lx + my + n = 0$; $px + qy + r = 0$ intersects the circle $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ at 4 points A and B ; C and D respectively.

If these points are concyclic then prove that $\begin{vmatrix} 2(g_1 - g_2) & 2(f_1 - f_2) & c_1 - c_2 \\ l & m & n \\ p & q & r \end{vmatrix} = 0$



Sol. The 3 radical axis are concurrent

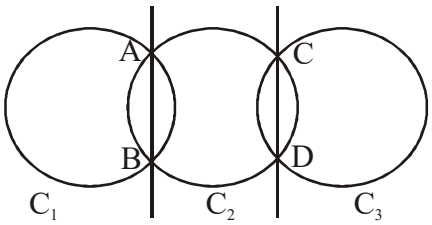
$lx + my + n = 0$

$px + qy + r = 0$

and $S_1 - S_2 = 0$

use condition of concurrency

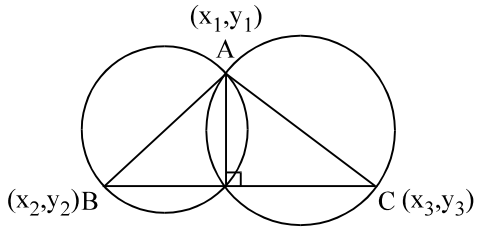
Alternate :



R.A. of C_2 & C_3 : L_2
R.A. of C_1 & C_3 : L_1
R.A. of C_1 & C_2 :
$2(g_1 - g_2)x + 2(f_1 - f_2)y + c_1 - c_2 = 0$

$$\Rightarrow \begin{vmatrix} 2(g_1 - g_2) & 2(f_1 - f_2) & (c_1 - c_2) \\ \ell & m & n \\ p & q & r \end{vmatrix} = 0$$

2. Prove analytically that radical centre of 3 circles described on the sides of a triangle as diameter is the orthocenter of the triangle. (Remember this)



Sol. (check slope of radical axes and BC) $(m_1m_2 = -1)$

$\Rightarrow$ Radical axes will act as an altitude.

Alternate :

C1 : $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

$RA \perp C_1C_2$

$C_1C_2 \parallel BC$

$RA \perp BC$

$AD \perp BC$

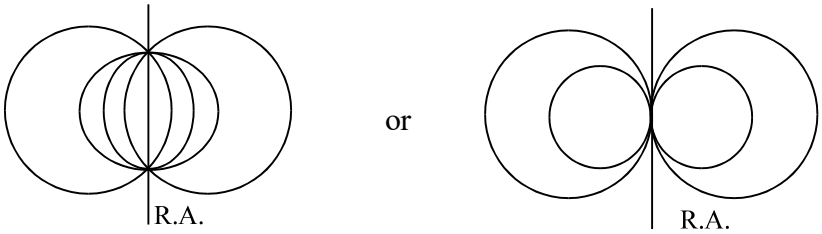
$\Rightarrow$ AD is altitude.

All radical axis will be altitude.

So, their P.O.I. will be radical center which will be orthocenter.

14.3 COAXIAL SYSTEM OF CIRCLES

Definition : A system of circles, every 2 of which have the same radical axis, is called Coaxial system of circles.



Questions on Coaxial system of circles

1. Find equation of circle passing through origin and coaxial with circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 6x + 3 = 0$.

Ans. $x^2 + y^2 - 9 + \lambda(x^2 + y^2 - 6x + 3) = 0$; $\lambda = 3$

Sol. $x^2 + y^2 - 9 + \lambda(x^2 + y^2 - 6x + 3) = 0$

passing through (0, 0).

$\Rightarrow 3\lambda - 9 = 0$

$\lambda = 3$

2. Find the equation of a circle coaxial with $S_1 : x^2 + y^2 + 4x + 2y + 1 = 0$ and $S_2 : 2x^2 + 2y^2 - 2x - 4y - 3 = 0$ and the centre of the circle to be determined lies on the radical axes of these 2 circle.

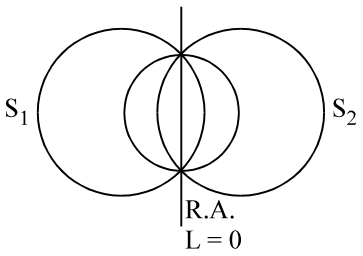
Ans $x^2 + y^2 + 4x + 2y + 1 + \lambda\left(5x + 4y + \frac{5}{2}\right) = 0; \lambda = \frac{23}{18}$

Sol. $S_1 : x^2 + y^2 + 4x + 2y + 1 = 0$

$S_2 : x^2 + y^2 - x - 2y - \frac{3}{2} = 0$

equation of radical axis $S_1 - S_2 = 0$

i.e. $5x + 4y + \frac{5}{2} = 0$



family of circles passes through $S_1 = 0$ and $L = 0$

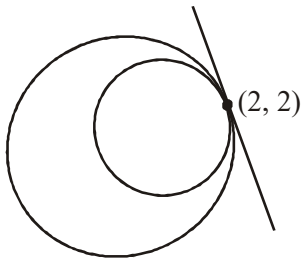
$S_1 + \lambda L = 0$ ....(1)

centre lies on $L = 0$, get $\lambda = \frac{23}{18}$

3. Find the equation of the circle passes through (1, 1) belonging to the system of coaxial circles which touches the locus of the point of intersection of 2 perpendiculars tangents to the circle $x^2 + y^2 = 4$ at (2, 2).

Ans. $(x-2)^2 + (y-2)^2 + \lambda(x+y-4) = 0 ; \lambda = 1$

Sol.

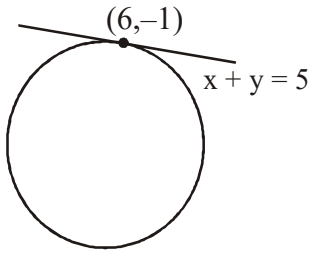


Equation of circle is :
$(x-2)^2 + (y-2)^2 + \lambda(x+y-4) = 0$
Passes through (1, 1)
$1 + 1 + \lambda(1 + 1 - 4) = 0$
$2 + 2\lambda = 0$
$\lambda = 1$

4. From a point P tangents drawn to the cirlces $x^2 + y^2 + x - 3 = 0$; $3x^2 + 3y^2 - 5x + 3y = 0$ and $4x^2 + 4y^2 + 8x + 7y + 9 = 0$ are of equal length. Find the equation of the circle passes through P and which touches the line $x + y = 5$ at (6, –1) **[O]**

Ans. $x^2 + y^2 - 7x + 7y + 12 = 0$

Sol.



Equation of circle is :
$\Rightarrow (x-6)^2 + (y+1)^2 + \lambda(x+y-5) = 0$
Passes through P.
Here, P is the radical centre.

$$\left.\begin{aligned} x^2 + y^2 + x - 3 = 0 \\ x^2 + y^2 - \frac{5}{3}x + y = 0 \end{aligned}\right] \xrightarrow{\text{R.A.}} \frac{8}{3}x - y - 3 = 0$$

$$\left.\begin{aligned} x^2 + y^2 - \frac{5}{3}x + y = 0 \\ x^2 + y^2 + 2x + \frac{7}{4}y + \frac{9}{4} = 0 \end{aligned}\right] \xrightarrow{\text{R.A.}} -\frac{11}{3}x - \frac{3}{4}y - \frac{9}{4} = 0$$

Radical centre P (0, –3)
$\Rightarrow 36 + 4 - 8\lambda = 0$

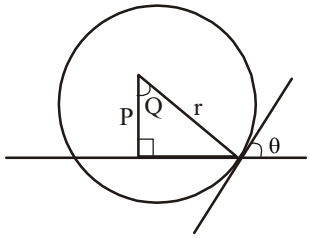
$\lambda = 5$

$\therefore$ Equation of circle is

$x^2 - 12x + 36 + y^2 + 1 + 2y + 5x + 5y - 25 = 0$

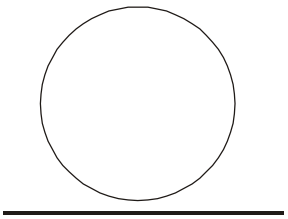
$x^2 + y^2 - 7x + 7y + 12 = 0$

15.1 ANGLE BETWEEN A LINE AND A CIRCLE



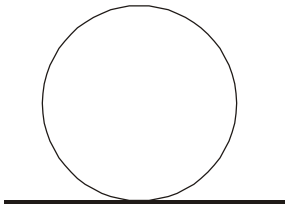
$\cos\theta = \frac{p}{r}$

Note – 1. If line & circle do not have a common point



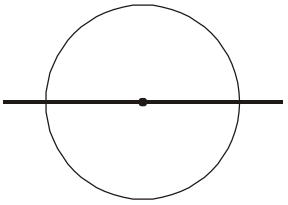
θ = not defined

2. If line touches the circle



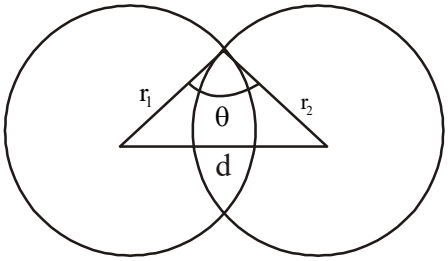
$\theta = 0^\circ$

3. If line cuts the circle orthogonally then it will pass through center.



15.2 ANGLE BETWEEN TWO CIRCLES AND ORTHOGONALITY OF TWO CIRCLES

The angle between the tangents of two circles at the point of intersection of the two circles is called angle of intersection of two circles.



$$\cos\theta = \left|\frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}\right|$$

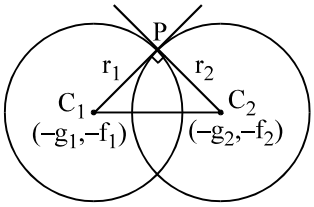
Orthogonality of two Circles

$\theta = 90^\circ$

$r_1^2 + r_1^2 = d^2$

$S_1 : x^2 + y^2 - 2g_1x + 2f_1y + c_1 = 0$

$S_2 : x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$



$r_1^2 + r_2^2 = (g_1 - g_2)^2 + (f_2 - f_2)^2$

$\Rightarrow \quad (g_1^2 + f_1^2 - c_1) + (g_2^2 + f_2^2 - c_2) = (g_1 - g_2)^2 + (f_1 - f_2)^2$

$\therefore \quad$ **$2 g_1 g_2 + 2 f_1 f_2 = c_1 + c_2$** (condition for orthogonality)

Note–

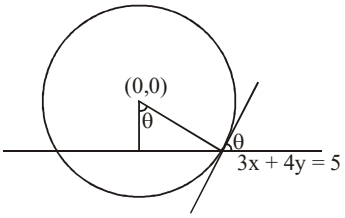
1. If two circles cut each other orthogonally then normal to 1 circle at common pt. will be tangent to other circle.
2. Locus of centre of a circle which cuts 2 given circles S_1 & S_2 orthogonally is radical axis of S_1 & S_2.

Questions on Orthogonality of circles

1. Find angle between $x^2 + y^2 = 2$ and $3x + 4y = 5$

Ans. $\theta = 45°$

Sol.



$P = \frac{5}{5} = 1, r = \sqrt{2}$

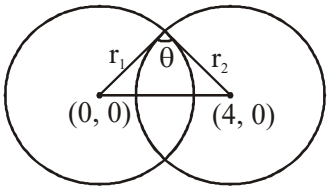
$\cos\theta = \frac{1}{\sqrt{2}}$

$\theta = 45^\circ$

2. Find angle between $x^2 + y^2 = 9$ and $x^2 + y^2 - 8x = 0$

Ans. $\theta = \cos^{-1}\frac{3}{8}$

Sol.



$r_1 = 3,\ r_2 = 4$

$d = 4$

$$\cos\theta = \left|\frac{9+16-16}{2\times3\times4}\right| = \frac{3}{8}$$

$$\theta = \cos^{-1}\left(\frac{3}{8}\right)$$

3. If the circles, $S_1 : x^2 + y^2 + 2x + 2ky + 6 = 0$ and $S_2 : x^2 + y^2 + 2kx + k = 0$ intersects orthogonally then find k.

Ans. $k = 6$

Sol. $$\cos 90^o = \left|\frac{r_1^2 + r_2^2 - d^2}{2r_1 - r_2}\right|$$

$\Rightarrow r_1^2 + r_2^2 = d^2$

$\Rightarrow 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

$2\cdot1\cdot k + 2\cdot k\cdot0 = 6 + k$

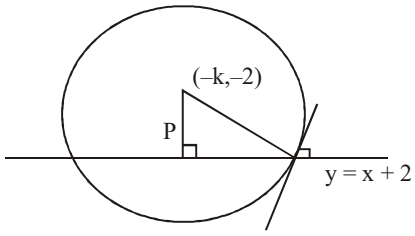
$2k = 6 + k$

$k = 6$

4. If line $y = x + 2$ is orthogonal to the circle $x^2 + y^2 + 2kx + 4y = 0$ then find k.

Ans. $k = 4$

Sol.



It is possible when $P = 0$

$|k - 4| = 0$

$k = 4$

5. If length of tangent from origin to the circle is 1 and lines $x = y$ and $x + y = 6$ intersect the circle orthogonally then find equation of circle.

Ans. $x^2 + y^2 - 6x - 6y + 1 = 0$

Sol. $\sqrt{S_1} = 1$

$\Rightarrow c = 1$

Given line $y = x$ & $x + y = 6$ intersects the circle orthogonally

$\Rightarrow$ these are Normal.

$\Rightarrow$ (3,3) will be centre $\Rightarrow r = \sqrt{9+9-1} = \sqrt{17}$

$\Rightarrow$ Equation of circle $x^2 + y^2 - 6x - 6y + 1 = 0$

6. Find equation of circle with centre at origin which cuts the circle $x^2 + y^2 + 12x + 4y + 16 = 0$ orthogonally.

Ans. $x^2 + y^2 = 16$

Sol. Let equation of circle : $x^2 + y^2 - c_1 = 0$

$\Rightarrow 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

$\Rightarrow 0 + 0 = -c_1 + 16$

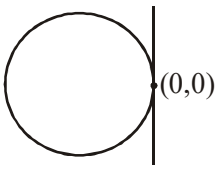
Equation $x^2 + y^2 = 16$

7. Find equation of circle touching the line $x + y = 0$ at origin and cuts the circle $x^2 + y^2 + 2x + 4x - 9 = 0$ orthogoanally.

ans. $x^2 + y^2 + \lambda(x+y) = 0$; $\lambda = -3$

Sol. Equation of circle :

$(x - 0)^2 + (y - 0)^2 + \lambda(x + y) = 0$



$\Rightarrow x^2 + y^2 + \lambda x + \lambda y = 0$ and $x^2 + y^2 + 2x + 4y - 9 = 0$

Cuts orthogonally

$$\Rightarrow 2\left(\frac{\lambda}{2}\right)\cdot 1 + 2\cdot\left(\frac{\lambda}{2}\right)\cdot 2 = -9 + 0$$

$\lambda = -3$

Equation of circle is

$x^2 + y^2 - 3x - 3y = 0$

8. Find equation of locus of centres of a circle which cuts the circle $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x + 3y = 0$ orthogonally.

Ans. $6x - 3y - 4 = 0$

Sol. Let co-ordinate of centre be (h, k) & equation of circle.

$x^2 + y^2 - 2hx - 2ky + c = 0$ ...(1)

(i) cuts $x^2 + y^2 = 4$ orthogonally

$2\cdot h\cdot 0 + 2\cdot k\cdot 0 = c - 4$

$\Rightarrow c = 4$

(ii) cuts $x^2 + y^2 - 6x + 3y = 0$ orthogonally

$$\Rightarrow 2\cdot h\cdot 3 + 2\cdot k\cdot\left(-\frac{3}{2}\right) = c + 0$$

$6h - 3k = 4$

$\Rightarrow 6x - 3y = 4$

9. $S_1 : x^2 + y^2 - 2x - 12 = 0$ $S_2 : x^2 + y^2 - 4x - 4 = 0$ $S_3 : x^2 + y^2 = 4$

Find equation of circle passing through points of intersection of S_1 and S_2 and cutting the circle S_3 orthogonally.

Ans. $S_1 + \lambda S_2 = 0$; $\lambda = -2$

Sol. Equation of circle :

$S_1 + \lambda S_2 = 0$

$\Rightarrow (x^2 + y^2 - 2x - 12) + \lambda(x^2 + y^2 - 4x - 4) = 0$

$(1 + \lambda)x^2 + (1 + \lambda)y^2 - \lambda(2 + 4\lambda)x - 12 - 4\lambda = 0$

$$x^2 + y^2 + \lambda\left(\frac{-2+4\lambda}{1+\lambda}\right)x + \left(\frac{-12-4\lambda}{1+\lambda}\right) = 0$$

Cuts S_3 orthogonally

$\Rightarrow 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

$$\Rightarrow 0 + 0 = \frac{-12-4\lambda}{1+\lambda} - 4$$

$$\frac{-12-4\lambda-4-4\lambda}{1+\lambda} = 0$$

$-8\lambda - 16 = 0$

$\lambda = -2$

10. Prove that the 2 circles which passes through the 2 points (0, a) and (0, – a) and touches the straight line y = mx + d will cut orthogonally if $d^2 = a^2(2 + m^2)$ **[O]**

Sol. Let the equation of circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$ ....(1)

$g \lessgtr \begin{matrix} g_1 \\ g_2 \end{matrix}$; $f \lessgtr \begin{matrix} f_1 \\ f_2 \end{matrix}$; $c \lessgtr \begin{matrix} f_1 \\ f_2 \end{matrix}$

Equation (1) passes through (0, a) and (0, – a)

$\therefore$ $a^2 + 2af + c = 0$

and $a^2 - 2af + c = 0$

add $2a^2 + 2c = 0$

$\therefore$ $a^2 = -c$

substitue $f = 0$

$\therefore$ $f_1 = f_2 = 0$

Now equation (1) becomes

$x^2 + y^2 + 2gx + c = 0$ ....(2)

line y = mx + d touches (2)

$$\therefore \quad \left|\frac{mg-d}{\sqrt{1+m^2}}\right| = \sqrt{g^2 - c}$$

$(mg - d)^2 = (g^2 - c)(1 + m^2)$

$m^2g^2 - 2mgd + d^2 = g^2 + m^2g^2 - c - m^2c$

$\begin{matrix} g_1 \\ g_2 \end{matrix} \gtrdot$ $g^2 + 2mgd - d^2 - c(1 + m^2) = 0$ ....(3)

Also $c = -a^2$ $\Rightarrow$ $c = c_1 = c_2 = -a^2$

using condition of orthogonality

$2g_1g_2 = 2c = -a^2$

$g_1g_2 = -a^2$

from equation (3) $g_1g_2 = -\{d^2 + c(1 + m^2)\}$

$\therefore$ $d^2 + c(1 + m^2) = a^2$

$c = -a^2$ $\Rightarrow$ $d^2 - a^2(1 + m^2) = a^2$

$\therefore$ $d^2 = a^2(1 + m^2)$ Hence proved.

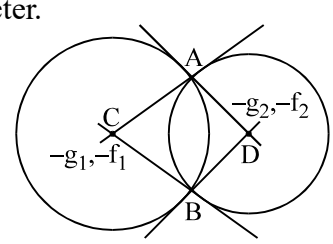
11. The circles $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ intersect orthogonally at A and B.

If C, D are the centres of these circles, show that the equation of the circle passing through A, B, C, D is $2(x^2 + y^2) + 2(g_1 + g_2)x + 2(f_1 + f_2)y + c_1 + c_2 = 0$. **[O]**

Sol. Since the circles intersect orthogonally hence the quad. ABCD is an cyclic quad.

$\therefore$ required equation of the circle will be the circle on CD are diameter.

$\therefore$ $(x + g_1)(x + g_2) + (y + f_1)(y + f_2) = 0$ ....(1)



Also condition of orthogonally gives

$2g_1 g_2 + 2f_1 f_2 = c_1 + c_2$ ....(2)

$\therefore$ $x^2 + y^2 + (g_1 + g_2)x + (f_1 + f_2)y + \frac{c_1 + c_2}{2} = 0$

To find equation of a circle cutting three circles orthogonally.

Ex.1 $S_1 : x^2 + y^2 = 1$
$S_2 : x^2 + y^2 + 8x + 15 = 0$
$S_3 : x^2 + y^2 - 12y + 35 = 0$
Find equation of circle which cuts S_1, S_2 & S_3 orthogonally.

Ans. $x^2 + y^2 + 4x - 6y + 1 = 0$

Sol. Let the equation of circle be : $x^2 + y^2 + 2gx + 2fy + c = 0$ ...(1)

(i) is orthogonal to S_1

$2g \cdot 0 + 2f \cdot 0 = c - 1$

$c = 1$

(i) is orthogonal to S_2

$2g \cdot 4 + 2f \cdot 0 = c + 15$

$8g = 16$

$g = 2$

(i) is orthogonal to S_3

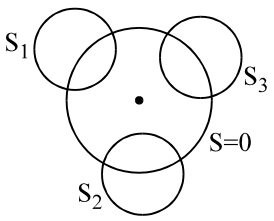
$2g \cdot 0 + 2f(-6) = c + 35$

$-12f = 36$

$f = -3$

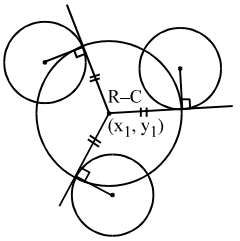
Equation of circle : $x^2 + y^2 + 4x - 6y + 1 = 0$

Method-I: Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$
orthogonal to $S_1 = 0$; $S_2 = 0$; $S_3 = 0$ get 3 relation in g, f, c.



Method-II : Circle orthogonal to 3 given circles is one such circle whose centre is the radical centre of 3 given circles and radius equals to length of tangent drawn from radical centre to any one of the circles.

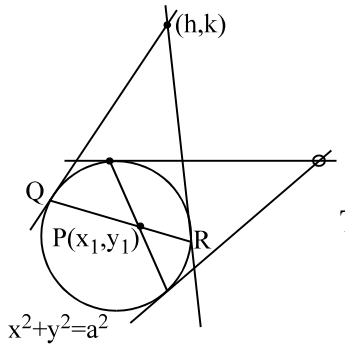
[Solve any 2 radical axis to get radical centre and then compute $r = L_T$]



16. POLE AND POLAR

If through a point P in the plane of the circle, there be drawn any straight line to meet the circle in Q

and R, the locus of the point of intersection of the tangents at Q & R is called the **Polar of the point P** and P is called the **Pole of the Polar**.



The equation of the polar is **T=0**

Note:

1. If point is outside the circle then equation of polar and chord of contact is same. So the chord of contact is polar.
2. If point is inside the circle then chord of contact does not exist but polar exists.
3. If point lies on the circle then polar and tangent on that point are same.

Questions on Pole & Polar **[O]**

1. Find the pole of a given line $lx + my = n$ w.r.t a circle $x^2 + y^2 = a^2$

Ans. $\left(\frac{a^2\ell}{n}, \frac{a^2m}{n}\right)$3

Sol

i.e we have to find the coordinates of the point whose polar w.r.t the given circle is the line $lx + my = n$

Let the pole be $P(x_1, y_1)$

$\therefore$ its polar w.r.t. the circle $x^2 + y^2 = a^2$ is

$$x x_1 + y y_1 = a^2 \ldots.(1)$$

also $l x + m y = n$ ....(2)

(1) and (2) must be identical

$$\therefore \quad \frac{a^2 l x_1}{n} = \frac{y_1}{m} = \frac{a^2}{n}$$

$$\therefore \quad x_1 = \frac{a^2 l}{n} \text{ and } y_1 = \frac{a^2 m}{n} \text{ (not to be remember)}$$

2. Find the value of 'k' for which the points (2, k) and (k, 3) are conjugate of each other w.r.t. the circle $x^2 + y^2 = 10$.

Ans 2

Sol. Polar of (2, k) is $2x + ky = 10$ ....(1)

passes through (k, 3)

$2k + 3k = 10 \Rightarrow k = 2$

3. The lengths of the tangents from two points A and B to a circle are l_1 and l_2 respectively. If the points are conjugate with respect to the circle $x^2 + y^2 = a^2$ then, show that $(AB)^2 = l_1^2 + l_2^2$.

Sol. Let $A(x_1, y_1)$ and $B(x_2, y_2)$.

Now $x_1^2 + y_1^2 - a^2 = l_1^2$ and $x_2^2 + y_2^2 - a^2 = l_2^2$.

Also $x x_1 + y y_1 = a^2$ contains (x_2, y_2)

Hence $x_1 x_2 + y_1 y_2 = a^2$. Now $(AB)^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 \Rightarrow$ result

4. Prove that if two lines at right angles are conjugate with respect to a circle, one of them must pass through the centre.

Sol. Let the line be $ax + by + c = 0$ ....(1)

and $bx - ay + d = 0$ ....(2)

now pole of (1) passes through (2) $\Rightarrow d = 0$

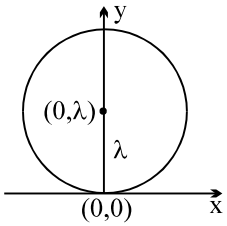
5. A variable circle is drawn to touch the axis of 'x' at origin. Find the locus of the pole of the straight line $lx + my + n = 0$ w.r.t. the circle.

Ans $nx + ly^2 = mxy$

Sol. Equation of the variable circle is

$(x - 0)^2 + (y - \lambda)^2 = \lambda^2$

$x^2 + y^2 - 2\lambda y = 0$ ....(1)



Let the pole of $lx + my + n = 0$ be (h, k)

$\therefore$ its polar w.r.t. (1) is

$hx + ky - \lambda(y + k) = 0$

i.e. $hx + y(k - \lambda) - \lambda k = 0$ ....(2)

Also polar is $lx + my + n = 0$ ....(3)

on comparing $\frac{h}{l} = \frac{k - \lambda}{m} = -\frac{\lambda k}{n}$

$\therefore$ $k - \lambda = \frac{mh}{l}$ and $\lambda = \frac{nh}{kl}$; $k - \frac{nh}{kl} = \frac{mh}{l}$ or $k^2 l - nh = mkh$

$\therefore$ required locus is $ly^2 = nx + mxy$

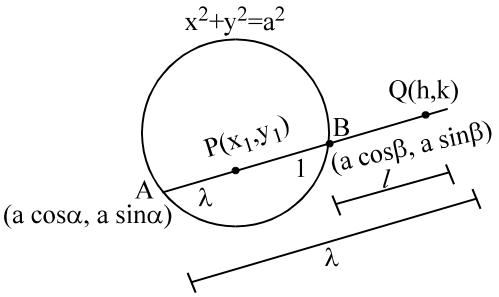
6. Prove that Locus of the harmonic conjugate of a given point $P(x_1, y_1)$ w.r.t. 2 given points in which any line through P cuts a given circle $x^2 + y^2 = a^2$ is polar of 'P' w.r.t. the given circle i.e. $x x_1 + y y_1 = a^2$.

Sol. $x_1 = \frac{a\lambda\cos\beta + a\cos\alpha}{\lambda + 1}$

$y_1 = \frac{a\lambda\sin\beta + a\sin\alpha}{\lambda + 1}$

$h = \frac{a\lambda\cos\beta - a\cos\alpha}{\lambda - 1}$

$k = \frac{a\lambda\sin\beta - a\sin\alpha}{\lambda - 1}$



$h x_1 + k y_1 = \frac{a^2 x^2 - a^2}{\lambda^2 - 1} = a^2$

$\therefore$ $x x_1 + y y_1 = a^2$ Hence proved.

7. Find the equation to the locus of the feet of the perpendicular drawn from the origin upon a variable chord of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which subtends right angle at origin.

Ans. $2x^2 + 2y^2 + 2gx + 2fy + c = 0$

Sol. Equation of chord AB with mid point M (h, k)

$y - k = -\frac{h}{k}(x - h)$

$ky - k^2 = -h^2 - hx$

$\therefore$ $hx + ky = h^2 + k^2$ ....(1)

Now homogenising $S = 0$ with the help of (1)

